Bus Bridge Server

commit-327ab86

Generated by Doxygen 1.9.8

Chapter 1

Bus Bridge Server

Chapter 2

Test List

File test_math.cpp

MathTest.Add

- Verifies that the `add` function correctly computes the sum of two integers.
- Example: `add(2, 3)` should return `5`.

MathTest.Subtract

- Verifies that the `subtract` function correctly computes the difference between two integers.
- Examples:
  - `subtract(10, 3)` should return `7`.
  - `subtract(9, 3)` should return `6`.

MathTest.SubtractNegative

- Verifies that the `subtract` function handles subtraction with negative integers correctly.
- Example: `subtract(10, -3)` should return `13`.

Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

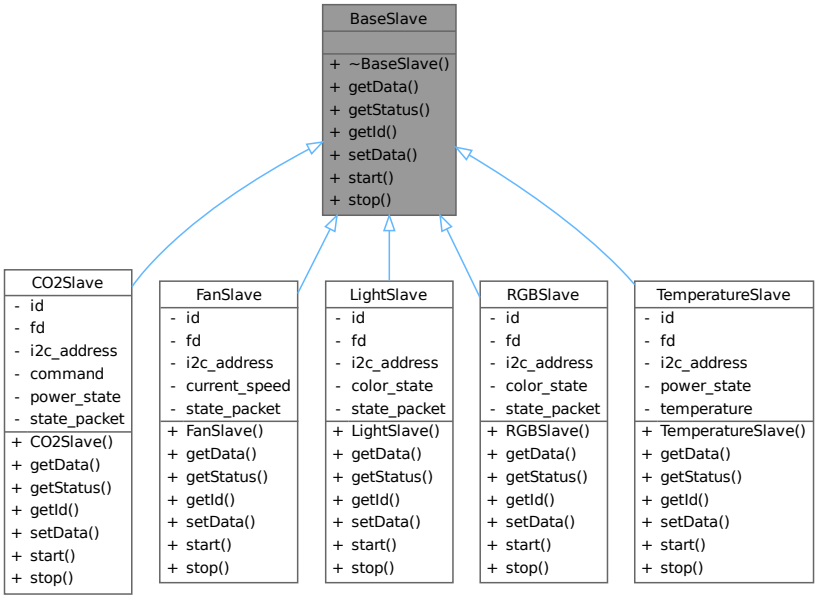
Chapter 6

Class Documentation

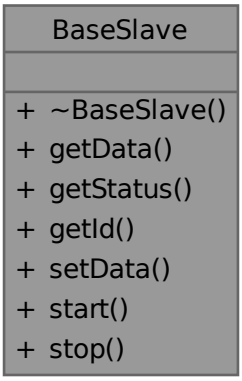
6.1 BaseSlave Class Reference

`#include <BaseSlave.h>`

Inheritance diagram for BaseSlave:



Collaboration diagram for BaseSlave:



Public Member Functions

- virtual ~BaseSlave ()=default
- virtual const void * getData ()=0
- virtual bool getStatus ()=0
- virtual int getId ()=0
- virtual void setData (void *data)=0
- virtual void start (int i2c_fd)=0
- virtual void stop ()=0

6.1.1 Detailed Description

Definition at line 6 of file BaseSlave.h.

6.1.2 Constructor & Destructor Documentation

6.1.2.1 ~BaseSlave()

```
virtual BaseSlave::~BaseSlave ( )  [virtual], [default]
```

6.1.3 Member Function Documentation

6.1.3.1 getData()

```
virtual const void * BaseSlave::getData ( )  [pure virtual]
```

Implemented in FanSlave, LightSlave, RGBSlave, TemperatureSlave, and CO2Slave.

6.1.3.2 getId()

```
virtual int BaseSlave::getId ( )  [pure virtual]
```

Implemented in FanSlave, LightSlave, RGBSlave, TemperatureSlave, and CO2Slave.

6.1.3.3 getStatus()

```
virtual bool BaseSlave::getStatus ( )  [pure virtual]
```

Implemented in FanSlave, LightSlave, RGBSlave, TemperatureSlave, and CO2Slave.

6.1.3.4 setData()

```
virtual void BaseSlave::setData (
            void * data )  [pure virtual]
```

Implemented in FanSlave, LightSlave, RGBSlave, TemperatureSlave, and CO2Slave.

6.1.3.5 start()

```
virtual void BaseSlave::start (
            int i2c_fd )  [pure virtual]
```

Implemented in FanSlave, LightSlave, RGBSlave, TemperatureSlave, and CO2Slave.

6.1.3.6 stop()

```
virtual void BaseSlave::stop ( )  [pure virtual]
```

Implemented in FanSlave, LightSlave, RGBSlave, TemperatureSlave, and CO2Slave.

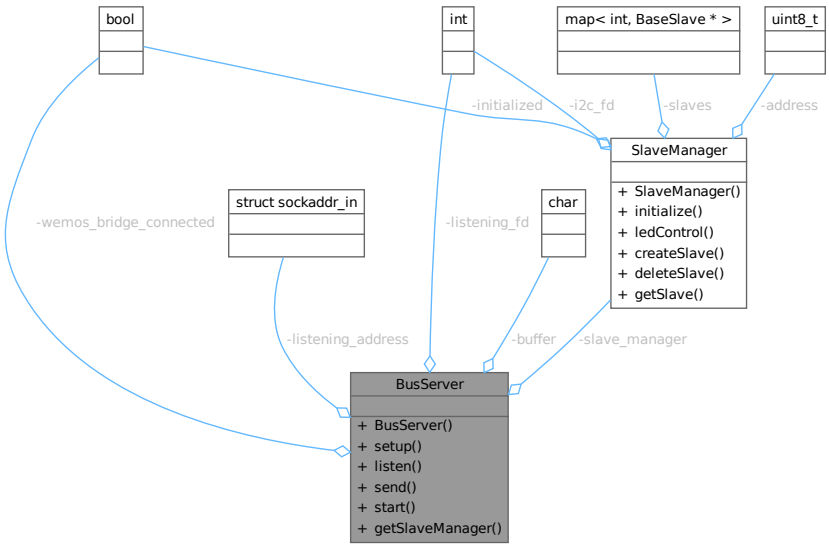
The documentation for this class was generated from the following file:

- include/BaseSlave.h

6.2 BusServer Class Reference

`#include <BusServer.h>`

Collaboration diagram for BusServer:



Public Member Functions

- BusServer ()
- void setup (std::string ip, int port)

  Setup for the IP socket and the I2C slaves connection.
- void listen ()

  Open the underlying socket for incoming connections.
- void send (struct sensor_packet *pkt, int fd)

  Send a sensor packet to a connected network client.
- void start ()

  Start the main loop of the BusServer.
- SlaveManager & getSlaveManager ()

  balls

Private Attributes

- int listening_fd
- struct sockaddr_in listening_address
- bool wemos_bridge_connected = false
- char buffer [BUFFER_SIZE]
- SlaveManager slave_manager

6.2.1 Detailed Description

Definition at line 12 of file BusServer.h.

6.2.2 Constructor & Destructor Documentation

6.2.2.1 BusServer()

```
BusServer::BusServer ( )  [inline]
```

Definition at line 14 of file BusServer.h.

6.2.3 Member Function Documentation

6.2.3.1 getSlaveManager()

```
SlaveManager & BusServer::getSlaveManager ( )
```

balls

Definition at line 179 of file BusServer.cpp.

6.2.3.2 listen()

```
void BusServer::listen ( )
```

Open the underlying socket for incoming connections.

Exceptions

std::runtime_error	if the listening fails

Definition at line 59 of file BusServer.cpp.

6.2.3.3 send()

```
void BusServer::send (
            struct sensor_packet * pkt,
            int fd )
```

Send a sensor packet to a connected network client.

Depending on the data set in the packet header, send a certain amount of data to the client.

Parameters

pkt	A pointer to the sensor_packet struct to send over
fd	The file descriptor to send the packet over

Exceptions

std::runtime_error	if the sending fails for any reason

Definition at line 66 of file BusServer.cpp.

6.2.3.4 setup()

```
void BusServer::setup (
            std::string ip,
            int port )
```

Setup for the IP socket and the I2C slaves connection.

This method will set up a socket for listening on the network and will also tell the underlying SlaveManager object to initialize its I2C bus

Parameters

ip	The IP address to listen on within the network
port	The TCP port to listen on

Exceptions

std::invalid_argument	if the passed IP address or port number are invalid
std::runtime_error	if the socket creation fails

Definition at line 19 of file BusServer.cpp.

6.2.3.5 start()

```
void BusServer::start ( )
```

Start the main loop of the BusServer.

This will first initialize the underlying I2C connections to the directly-connected slave devices, and then start accepting and processing network clients

Definition at line 76 of file BusServer.cpp.

6.2.4 Member Data Documentation

6.2.4.1 buffer

```
char BusServer::buffer[BUFFER_SIZE]  [private]
```

Definition at line 60 of file BusServer.h.

6.2.4.2 listening_address

```
struct sockaddr_in BusServer::listening_address  [private]
```

Definition at line 58 of file BusServer.h.

6.2.4.3 listening_fd

```
int BusServer::listening_fd  [private]
```

Definition at line 56 of file BusServer.h.

6.2.4.4 slave_manager

```
SlaveManager BusServer::slave_manager  [private]
```

Definition at line 62 of file BusServer.h.

6.2.4.5 wemos_bridge_connected

```
bool BusServer::wemos_bridge_connected = false  [private]
```

Definition at line 59 of file BusServer.h.

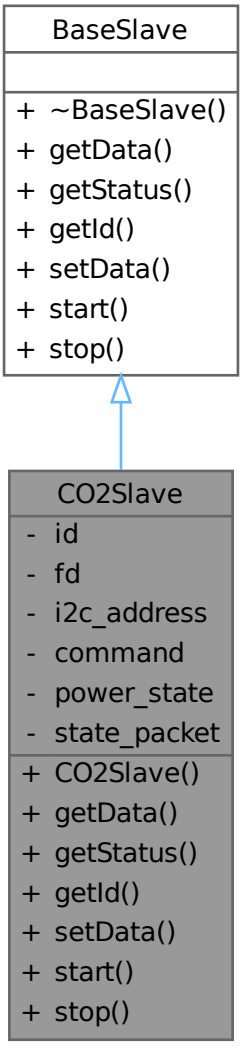
The documentation for this class was generated from the following files:

- include/BusServer.h
- src/BusServer.cpp

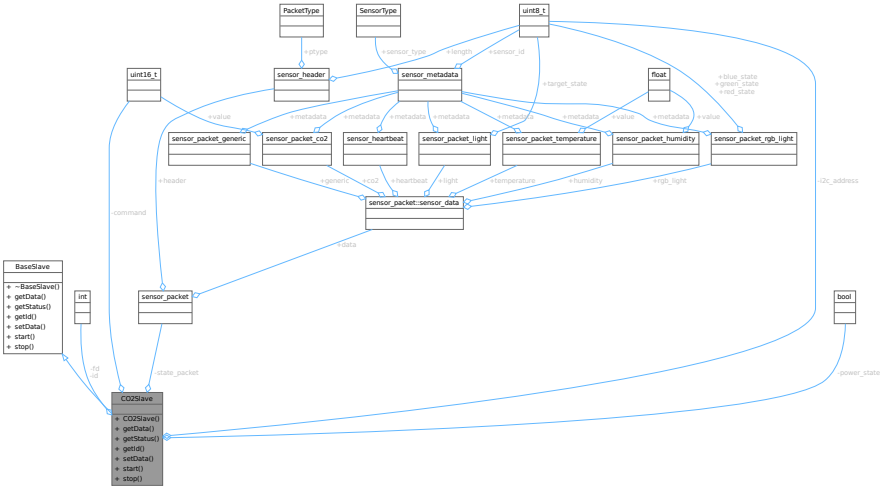
6.3 CO2Slave Class Reference

```
#include <CO2Slave.h>
```

Inheritance diagram for CO2Slave:



Collaboration diagram for CO2Slave:



Public Member Functions

- CO2Slave (uint8_t id, uint8_t i2c_address)
- const void * getData () override
- bool getStatus () override
- int getId () override
- void setData (void *data) override
- void start (int i2c_fd) override
- void stop () override

Public Member Functions inherited from BaseSlave

- virtual ~BaseSlave ()=default

Private Attributes

- int id
- int fd
- uint8_t i2c_address
- uint16_t command
- bool power_state = true
- struct sensor_packet state_packet

6.3.1 Detailed Description

Definition at line 7 of file CO2Slave.h.

6.3.2 Constructor & Destructor Documentation

6.3.2.1 CO2Slave()

```
CO2Slave::CO2Slave (
            uint8_t id,
            uint8_t i2c_address )
```

Definition at line 5 of file CO2Slave.cpp.

6.3.3 Member Function Documentation

6.3.3.1 getData()

```
const void * CO2Slave::getData ( )  [override], [virtual]
```

Implements BaseSlave.

Definition at line 18 of file CO2Slave.cpp.

6.3.3.2 getId()

```
int CO2Slave::getId ( )  [override], [virtual]
```

Implements BaseSlave.

Definition at line 33 of file CO2Slave.cpp.

6.3.3.3 getStatus()

```
bool CO2Slave::getStatus ( )  [override], [virtual]
```

Implements BaseSlave.

Definition at line 31 of file CO2Slave.cpp.

6.3.3.4 setData()

```
void CO2Slave::setData (
            void * data )  [override], [virtual]
```

Implements BaseSlave.

Definition at line 35 of file CO2Slave.cpp.

6.3.3.5 start()

```
void CO2Slave::start (
            int i2c_fd )  [override], [virtual]
```

Implements BaseSlave.

Definition at line 37 of file CO2Slave.cpp.

6.3.3.6 stop()

```
void CO2Slave::stop ( )  [override], [virtual]
```

Implements BaseSlave.

Definition at line 39 of file CO2Slave.cpp.

6.3.4 Member Data Documentation

6.3.4.1 command

```
uint16_t CO2Slave::command  [private]
```

Definition at line 22 of file CO2Slave.h.

6.3.4.2 fd

```
int CO2Slave::fd  [private]
```

Definition at line 20 of file CO2Slave.h.

6.3.4.3 i2c_address

```
uint8_t CO2Slave::i2c_address  [private]
```

Definition at line 21 of file CO2Slave.h.

6.3.4.4 id

```
int CO2Slave::id  [private]
```

Definition at line 19 of file CO2Slave.h.

6.3.4.5 power_state

```
bool CO2Slave::power_state = true  [private]
```

Definition at line 24 of file CO2Slave.h.

6.3.4.6 state_packet

```
struct sensor_packet CO2Slave::state_packet  [private]
```

Definition at line 25 of file CO2Slave.h.

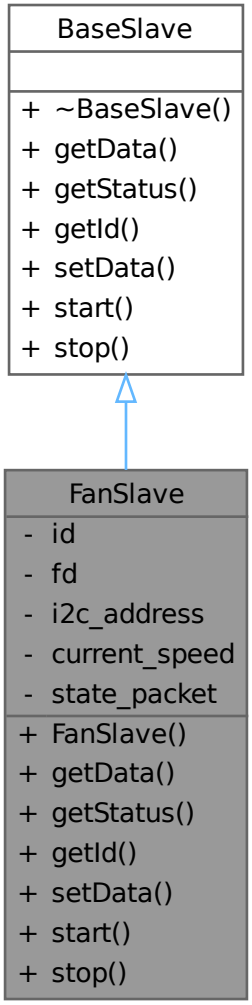
The documentation for this class was generated from the following files:

- include/CO2Slave.h
- src/CO2Slave.cpp

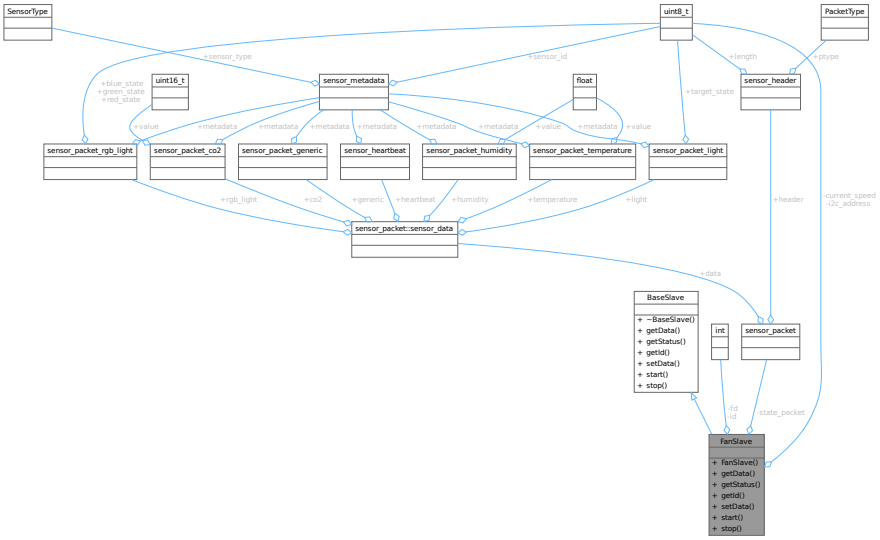
6.4 FanSlave Class Reference

```
#include <FanSlave.h>
```

Inheritance diagram for FanSlave:



Collaboration diagram for FanSlave:



Public Member Functions

- FanSlave (uint8_t id, uint8_t i2c_address)
- const void * getData ()
- bool getStatus ()
- int getId ()
- void setData (void *data)
- void start (int i2c_fd)
- void stop ()

Public Member Functions inherited from BaseSlave

- virtual ~BaseSlave ()=default

Private Attributes

- int id
- int fd
- uint8_t i2c_address
- FanData current_speed
- struct sensor_packet state_packet

6.4.1 Detailed Description

Definition at line 10 of file FanSlave.h.

6.4.2 Constructor & Destructor Documentation

6.4.2.1 FanSlave()

```
FanSlave::FanSlave (
            uint8_t id,
            uint8_t i2c_address )
```

Definition at line 3 of file FanSlave.cpp.

6.4.3 Member Function Documentation

6.4.3.1 getData()

```
const void * FanSlave::getData ( )  [virtual]
```

Implements BaseSlave.

Definition at line 5 of file FanSlave.cpp.

6.4.3.2 getId()

```
int FanSlave::getId ( )  [virtual]
```

Implements BaseSlave.

Definition at line 15 of file FanSlave.cpp.

6.4.3.3 getStatus()

```
bool FanSlave::getStatus ( )  [virtual]
```

Implements BaseSlave.

Definition at line 7 of file FanSlave.cpp.

6.4.3.4 setData()

```
void FanSlave::setData (
            void * data )  [virtual]
```

Implements BaseSlave.

Definition at line 17 of file FanSlave.cpp.

6.4.3.5 start()

```
void FanSlave::start (
            int i2c_fd )  [virtual]
```

Implements BaseSlave.

Definition at line 23 of file FanSlave.cpp.

6.4.3.6 stop()

```
void FanSlave::stop ( )  [virtual]
```

Implements BaseSlave.

Definition at line 25 of file FanSlave.cpp.

6.4.4 Member Data Documentation

6.4.4.1 current_speed

```
FanData FanSlave::current_speed  [private]
```

Definition at line 25 of file FanSlave.h.

6.4.4.2 fd

```
int FanSlave::fd  [private]
```

Definition at line 22 of file FanSlave.h.

6.4.4.3 i2c_address

```
uint8_t FanSlave::i2c_address  [private]
```

Definition at line 23 of file FanSlave.h.

6.4.4.4 id

```
int FanSlave::id  [private]
```

Definition at line 21 of file FanSlave.h.

6.4.4.5 state_packet

```
struct sensor_packet FanSlave::state_packet  [private]
```

Definition at line 27 of file FanSlave.h.

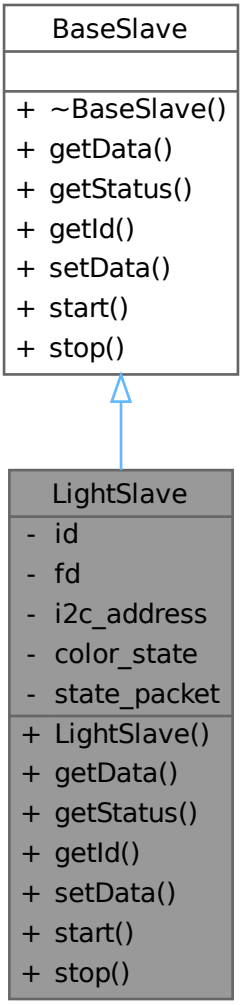
The documentation for this class was generated from the following files:

- include/FanSlave.h
- src/FanSlave.cpp

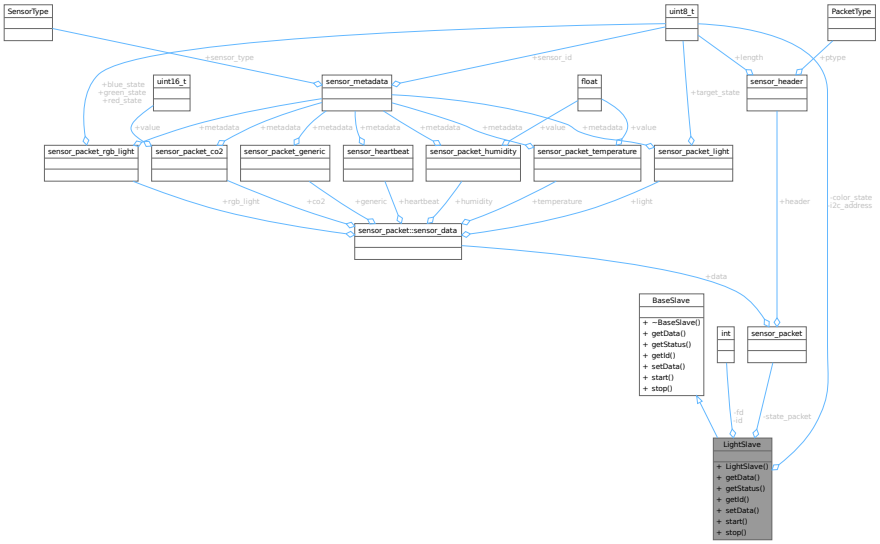
6.5 LightSlave Class Reference

```
#include <LightSlave.h>
```

Inheritance diagram for LightSlave:



Collaboration diagram for LightSlave:



Public Member Functions

- LightSlave (uint8_t id, uint8_t i2c_address)
- const void * getData ()
- bool getStatus ()
- int getId ()
- void setData (void *data)
- void start (int i2c_fd)
- void stop ()

Public Member Functions inherited from BaseSlave

- virtual ~BaseSlave ()=default

Private Attributes

- int id
- int fd
- uint8_t i2c_address
- LightData color_state
- struct sensor_packet state_packet

6.5.1 Detailed Description

Definition at line 9 of file LightSlave.h.

6.5.2 Constructor & Destructor Documentation

6.5.2.1 LightSlave()

```
LightSlave::LightSlave (
            uint8_t id,
            uint8_t i2c_address )
```

Definition at line 7 of file LightSlave.cpp.

6.5.3 Member Function Documentation

6.5.3.1 getData()

```
const void * LightSlave::getData ( )  [virtual]
```

Implements BaseSlave.

Definition at line 15 of file LightSlave.cpp.

6.5.3.2 getId()

```
int LightSlave::getId ( )  [virtual]
```

Implements BaseSlave.

Definition at line 28 of file LightSlave.cpp.

6.5.3.3 getStatus()

```
bool LightSlave::getStatus ( )  [virtual]
```

Implements BaseSlave.

Definition at line 23 of file LightSlave.cpp.

6.5.3.4 setData()

```
void LightSlave::setData (
            void * data )  [virtual]
```

Implements BaseSlave.

Definition at line 30 of file LightSlave.cpp.

6.5.3.5 start()

```
void LightSlave::start (
            int i2c_fd )  [virtual]
```

Implements BaseSlave.

Definition at line 39 of file LightSlave.cpp.

6.5.3.6 stop()

```
void LightSlave::stop ( )  [virtual]
```

Implements BaseSlave.

Definition at line 41 of file LightSlave.cpp.

6.5.4 Member Data Documentation

6.5.4.1 color_state

```
LightData LightSlave::color_state  [private]
```

Definition at line 24 of file LightSlave.h.

6.5.4.2 fd

```
int LightSlave::fd  [private]
```

Definition at line 21 of file LightSlave.h.

6.5.4.3 i2c_address

```
uint8_t LightSlave::i2c_address  [private]
```

Definition at line 22 of file LightSlave.h.

6.5.4.4 id

```
int LightSlave::id  [private]
```

Definition at line 20 of file LightSlave.h.

6.5.4.5 state_packet

```
struct sensor_packet LightSlave::state_packet  [private]
```

Definition at line 26 of file LightSlave.h.

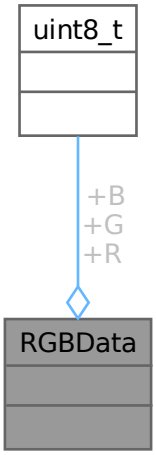
The documentation for this class was generated from the following files:

- include/LightSlave.h
- src/LightSlave.cpp

6.6 RGBData Struct Reference

```
#include <RGBSlave.h>
```

Collaboration diagram for RGBData:



Public Attributes

- uint8_t R
- uint8_t G
- uint8_t B

6.6.1 Detailed Description

Definition at line 7 of file RGBSlave.h.

6.6.2 Member Data Documentation

6.6.2.1 B

```
uint8_t RGBData::B
```

Definition at line 8 of file RGBSlave.h.

6.6.2.2 G

```
uint8_t RGBData::G
```

Definition at line 8 of file RGBSlave.h.

6.6.2.3 R

```
uint8_t RGBData::R
```

Definition at line 8 of file RGBSlave.h.

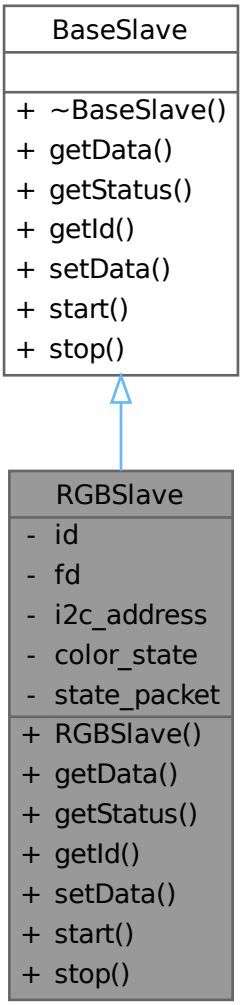
The documentation for this struct was generated from the following file:

- include/RGBSlave.h

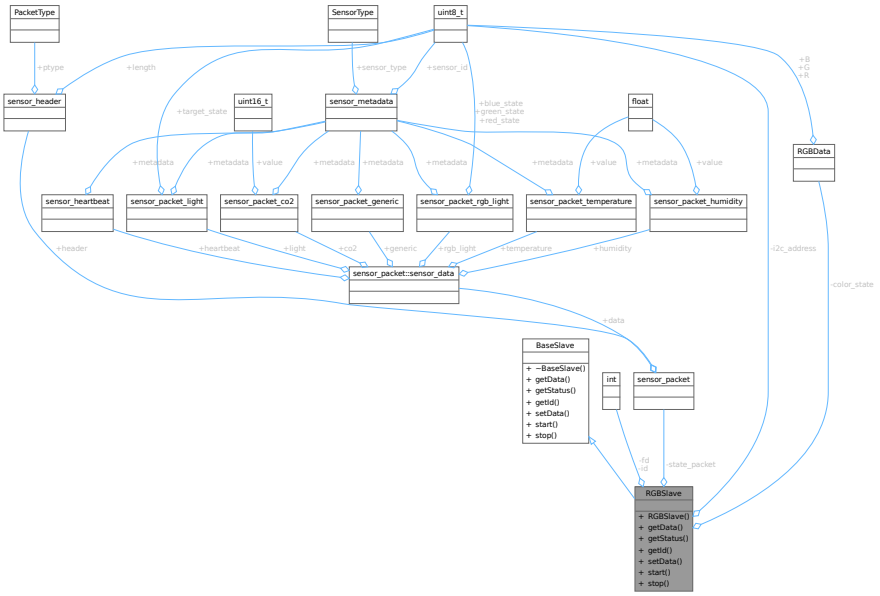
6.7 RGBSlave Class Reference

```
#include <RGBSlave.h>
```

Inheritance diagram for RGBSlave:



Collaboration diagram for RGBSlave:



Public Member Functions

- RGBSlave (uint8_t id, uint8_t i2c_address)
- const void * getData ()
- bool getStatus ()
- int getId ()
- void setData (void *data)
- void start (int i2c_fd)
- void stop ()

Public Member Functions inherited from BaseSlave

- virtual ~BaseSlave ()=default

Private Attributes

- int id
- int fd
- uint8_t i2c_address
- RGBData color_state
- struct sensor_packet state_packet

6.7.1 Detailed Description

Definition at line 11 of file RGBSlave.h.

6.7.2 Constructor & Destructor Documentation

6.7.2.1 RGBSlave()

```
RGBSlave::RGBSlave (
            uint8_t id,
            uint8_t i2c_address )
```

Definition at line 7 of file RGBSlave.cpp.

6.7.3 Member Function Documentation

6.7.3.1 getData()

```
const void * RGBSlave::getData ( )  [virtual]
```

Implements BaseSlave.

Definition at line 15 of file RGBSlave.cpp.

6.7.3.2 getId()

```
int RGBSlave::getId ( )  [virtual]
```

Implements BaseSlave.

Definition at line 42 of file RGBSlave.cpp.

6.7.3.3 getStatus()

```
bool RGBSlave::getStatus ( )  [virtual]
```

Implements BaseSlave.

Definition at line 33 of file RGBSlave.cpp.

6.7.3.4 setData()

```
void RGBSlave::setData (
            void * data )  [virtual]
```

Implements BaseSlave.

Definition at line 44 of file RGBSlave.cpp.

6.7.3.5 start()

```
void RGBSlave::start (
            int i2c_fd )  [virtual]
```

Implements BaseSlave.

Definition at line 65 of file RGBSlave.cpp.

6.7.3.6 stop()

```
void RGBSlave::stop ( )  [virtual]
```

Implements BaseSlave.

Definition at line 67 of file RGBSlave.cpp.

6.7.4 Member Data Documentation

6.7.4.1 color_state

```
RGBData RGBSlave::color_state  [private]
```

Definition at line 26 of file RGBSlave.h.

6.7.4.2 fd

```
int RGBSlave::fd  [private]
```

Definition at line 23 of file RGBSlave.h.

6.7.4.3 i2c_address

```
uint8_t RGBSlave::i2c_address  [private]
```

Definition at line 24 of file RGBSlave.h.

6.7.4.4 id

```
int RGBSlave::id  [private]
```

Definition at line 22 of file RGBSlave.h.

6.7.4.5 state_packet

```
struct sensor_packet RGBSlave::state_packet  [private]
```

Definition at line 28 of file RGBSlave.h.

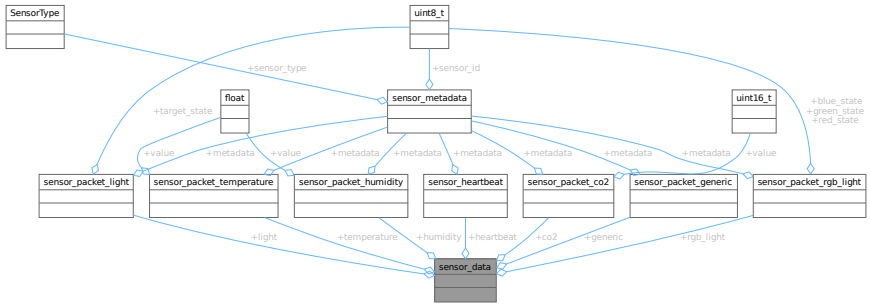
The documentation for this class was generated from the following files:

- include/RGBSlave.h
- src/RGBSlave.cpp

6.8 sensor_data Union Reference

```
#include <packets.h>
```

Collaboration diagram for sensor_data:



Public Attributes

- struct sensor_heartbeat heartbeat
- struct sensor_packet_generic generic
- struct sensor_packet_temperature temperature
- struct sensor_packet_co2 co2
- struct sensor_packet_humidity humidity
- struct sensor_packet_light light
- struct sensor_packet_rgb_light rgb_light

6.8.1 Detailed Description

Definition at line 4 of file packets.h.

6.8.2 Member Data Documentation

6.8.2.1 co2

```
struct sensor_packet_co2 sensor_data::co2
```

Definition at line 8 of file packets.h.

6.8.2.2 generic

```
struct sensor_packet_generic sensor_data::generic
```

Definition at line 6 of file packets.h.

6.8.2.3 heartbeat

```
struct sensor_heartbeat sensor_data::heartbeat
```

Definition at line 5 of file packets.h.

6.8.2.4 humidity

```
struct sensor_packet_humidity sensor_data::humidity
```

Definition at line 9 of file packets.h.

6.8.2.5 light

```
struct sensor_packet_light sensor_data::light
```

Definition at line 10 of file packets.h.

6.8.2.6 rgb_light

```
struct sensor_packet_rgb_light sensor_data::rgb_light
```

Definition at line 11 of file packets.h.

6.8.2.7 temperature

```
struct sensor_packet_temperature sensor_data::temperature
```

Definition at line 7 of file packets.h.

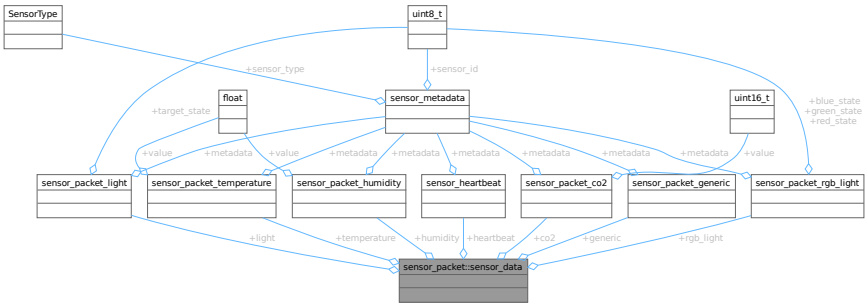
The documentation for this union was generated from the following file:

- include/packets.h

6.9 sensor_packet::sensor_data Union Reference

`#include <packets.h>`

Collaboration diagram for sensor_packet::sensor_data:



Public Attributes

- struct sensor_heartbeat heartbeat
- struct sensor_packet_generic generic
- struct sensor_packet_temperature temperature
- struct sensor_packet_co2 co2
- struct sensor_packet_humidity humidity
- struct sensor_packet_light light
- struct sensor_packet_rgb_light rgb_light

6.9.1 Detailed Description

Definition at line 240 of file packets.h.

6.9.2 Member Data Documentation

6.9.2.1 co2

`struct sensor_packet_co2 sensor_packet::sensor_data::co2`

Definition at line 244 of file packets.h.

6.9.2.2 generic

`struct sensor_packet_generic sensor_packet::sensor_data::generic`

Definition at line 242 of file packets.h.

6.9.2.3 heartbeat

```
struct sensor_heartbeat sensor_packet::sensor_data::heartbeat
```

Definition at line 241 of file packets.h.

6.9.2.4 humidity

```
struct sensor_packet_humidity sensor_packet::sensor_data::humidity
```

Definition at line 245 of file packets.h.

6.9.2.5 light

```
struct sensor_packet_light sensor_packet::sensor_data::light
```

Definition at line 246 of file packets.h.

6.9.2.6 rgb_light

```
struct sensor_packet_rgb_light sensor_packet::sensor_data::rgb_light
```

Definition at line 247 of file packets.h.

6.9.2.7 temperature

```
struct sensor_packet_temperature sensor_packet::sensor_data::temperature
```

Definition at line 243 of file packets.h.

The documentation for this union was generated from the following file:

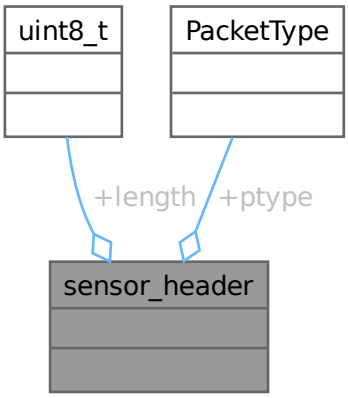
- include/packets.h

6.10 sensor_header Struct Reference

Header structure for sensor packets.

`#include <packets.h>`

Collaboration diagram for sensor_header:



Public Attributes

- uint8_t length

  Length of the packet excluding the header.
- PacketType ptype

  Type of the packet as PacketType (DATA, HEARTBEAT, etc.).

6.10.1 Detailed Description

Header structure for sensor packets.

Definition at line 41 of file packets.h.

6.10.2 Member Data Documentation

6.10.2.1 length

`uint8_t sensor_header::length`

Length of the packet excluding the header.

Definition at line 43 of file packets.h.

6.10.2.2 ptype

```
PacketType sensor_header::ptype
```

Type of the packet as PacketType (DATA, HEARTBEAT, etc.).

Definition at line 45 of file packets.h.

The documentation for this struct was generated from the following file:

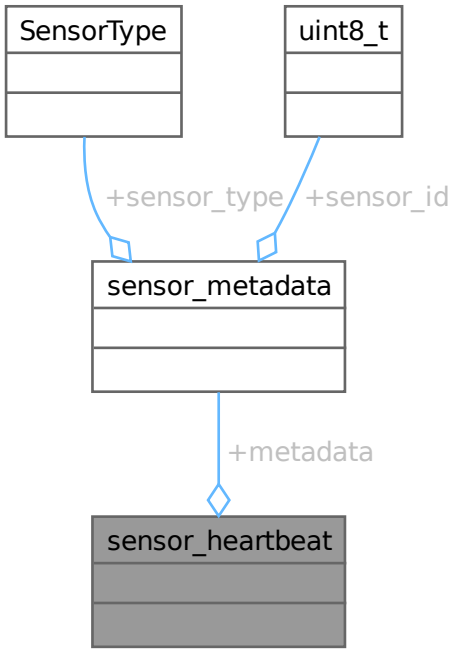
- include/packets.h

6.11 sensor_heartbeat Struct Reference

Structure for heartbeat packets.

```
#include <packets.h>
```

Collaboration diagram for sensor_heartbeat:



Public Attributes

- struct sensor_metadata metadata

6.11.1 Detailed Description

Structure for heartbeat packets.

This structure contains the type and ID of the sensor being addressed. This structure is used for heartbeat packets sent by the sensors to indicate they are still alive.

Definition at line 70 of file packets.h.

6.11.2 Member Data Documentation

6.11.2.1 metadata

```
struct sensor_metadata sensor_heartbeat::metadata
```

Definition at line 71 of file packets.h.

The documentation for this struct was generated from the following file:

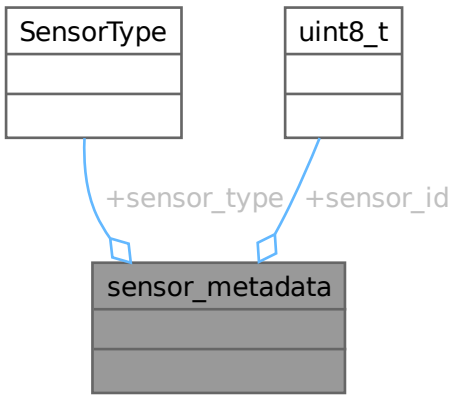
- include/packets.h

6.12 sensor_metadata Struct Reference

Structure for sensor metadata, which is always included in any packet.

```
#include <packets.h>
```

Collaboration diagram for sensor_metadata:



Public Attributes

- SensorType sensor_type
  Type of the sensor being addressed as SensorType (one byte)
- uint8_t sensor_id
  ID of the sensor being addressed.

6.12.1 Detailed Description

Structure for sensor metadata, which is always included in any packet.

Definition at line 53 of file packets.h.

6.12.2 Member Data Documentation

6.12.2.1 sensor_id

```
uint8_t sensor_metadata::sensor_id
```

ID of the sensor being addressed.

Definition at line 57 of file packets.h.

6.12.2.2 sensor_type

```
SensorType sensor_metadata::sensor_type
```

Type of the sensor being addressed as SensorType (one byte)

Definition at line 55 of file packets.h.

The documentation for this struct was generated from the following file:

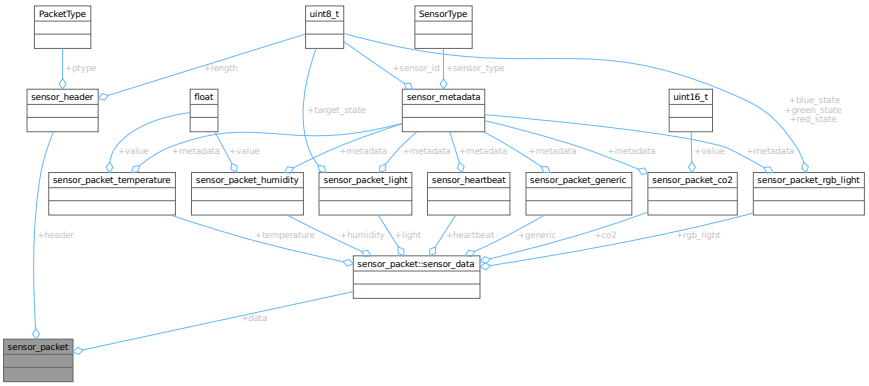
- include/packets.h

6.13 sensor_packet Struct Reference

Union structure for the entire sensor packet.

```
#include <packets.h>
```

Collaboration diagram for sensor_packet:



Classes

- union sensor_data

Public Attributes

- struct sensor_header header

  Header of the packet containing length and type information.
- union sensor_packet::sensor_data data

6.13.1 Detailed Description

Union structure for the entire sensor packet.

This structure is used to encapsulate the different types of sensor packets that can be sent and has the shape of a valid packet.

It contains a sensor_header followed by a union of different sensor data types. The union allows for different types of sensor data to be stored in the same memory location, depending on the packet type.

Example usage:
```
sensor_packet packet;
packet.header.length = sizeof(sensor_packet_generic);
packet.header.ptype = PacketType::DATA;
packet.data.generic.metadata.sensor_type = SensorType::BUTTON;
packet.data.generic.metadata.sensor_id = 1;

// Accessing the packet data
if (packet.header.ptype == PacketType::DATA) {
    if (packet.data.generic.metadata.sensor_type == SensorType::BUTTON) {
        uint8_t sensor_id = packet.data.generic.metadata.sensor_id;
        // Process button press event for sensor_id
    }
}
```

To use this structure to request data from the dashboard, you can set the ptype to DASHBOARD_GET to indicate that you want to request data from the backend (wemos bridge). Then, you use a sensor_packet_generic to specify the type of sensor you want to request data for and the ID of that sensor.

Example: We want to request temperature data from the backend (wemos bridge) for sensor ID 1.
```
sensor_packet packet;
packet.header.length = sizeof(sensor_packet_generic);
packet.header.ptype = PacketType::DASHBOARD_GET;
packet.data.generic.metadata.sensor_type = SensorType::TEMPERATURE;
packet.data.generic.metadata.sensor_id = 1;
```

The backend (wemos bridge) will then respond with a packet of type DASHBOARD_RESPONSE containing the requested data. Following the correct type packet for this example would be a sensor_packet_temperature.

Example: We want to change the color of an RGB light with ID 1 to red (255, 0, 0).
```
sensor_packet packet;
packet.header.length = sizeof(sensor_packet_rgb_light);
packet.header.ptype = PacketType::DASHBOARD_POST;
packet.data.rgb_light.metadata.sensor_type = SensorType::RGB_LIGHT;
packet.data.rgb_light.metadata.sensor_id = 1;
packet.data.rgb_light.red_state = 255;
packet.data.rgb_light.green_state = 0;
packet.data.rgb_light.blue_state = 0;
```

Note

The data field is a union that can hold different types of sensor data.

Definition at line 235 of file packets.h.

6.13.2 Member Data Documentation

6.13.2.1 data

```
union sensor_packet::sensor_data sensor_packet::data
```

6.13.2.2 header

```
struct sensor_header sensor_packet::header
```

Header of the packet containing length and type information.

Definition at line 237 of file packets.h.

The documentation for this struct was generated from the following file:

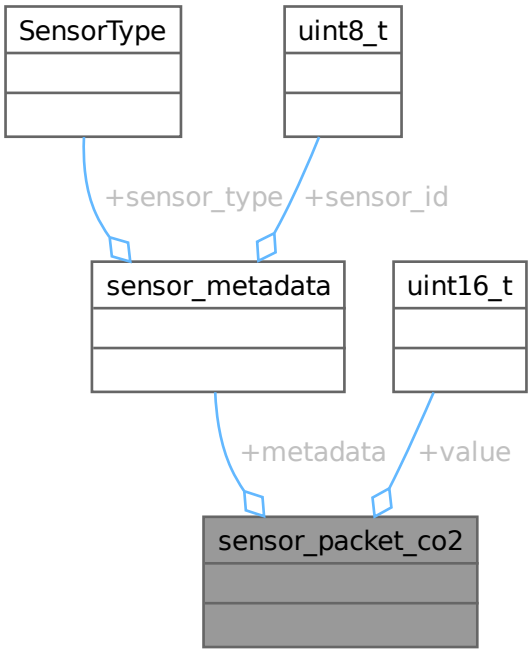
- include/packets.h

6.14 sensor_packet_co2 Struct Reference

Structure for CO2 sensor packets.

```
#include <packets.h>
```

Collaboration diagram for sensor_packet_co2:



Public Attributes

- struct sensor_metadata metadata
- uint16_t value

  Value of the sensor reading the CO2 level represented in ppm.

6.14.1 Detailed Description

Structure for CO2 sensor packets.

This structure contains the type, ID, and value of the CO2 sensor reading.

Note

The CO2 value is represented in parts per million (ppm).

Definition at line 108 of file packets.h.

6.14.2 Member Data Documentation

6.14.2.1 metadata

```
struct sensor_metadata sensor_packet_co2::metadata
```

Definition at line 109 of file packets.h.

6.14.2.2 value

```
uint16_t sensor_packet_co2::value
```

Value of the sensor reading the CO2 level represented in ppm.

Definition at line 111 of file packets.h.

The documentation for this struct was generated from the following file:

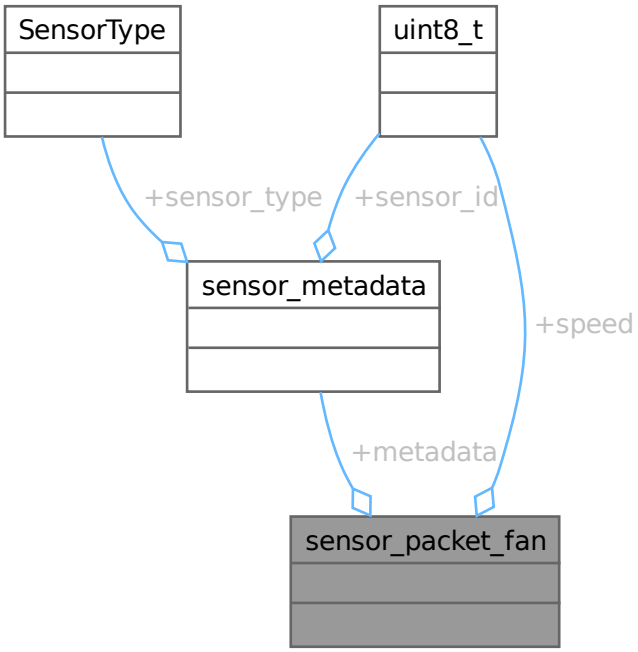
- include/packets.h

6.15 sensor_packet_fan Struct Reference

Structure for packets intended to control the fan speed.

`#include <packets.h>`

Collaboration diagram for sensor_packet_fan:



Public Attributes

- struct sensor_metadata metadata
- uint8_t speed

  Target speed of the fan (0 -> 255 = 0 -> 100%)

6.15.1 Detailed Description

Structure for packets intended to control the fan speed.

Send the fan speed (0 -> 255) using this packet

Definition at line 164 of file packets.h.

6.15.2 Member Data Documentation

6.15.2.1 metadata

```
struct sensor_metadata sensor_packet_fan::metadata
```

Definition at line 165 of file packets.h.

6.15.2.2 speed

```
uint8_t sensor_packet_fan::speed
```

Target speed of the fan (0 -> 255 = 0 -> 100%)

Definition at line 167 of file packets.h.

The documentation for this struct was generated from the following file:

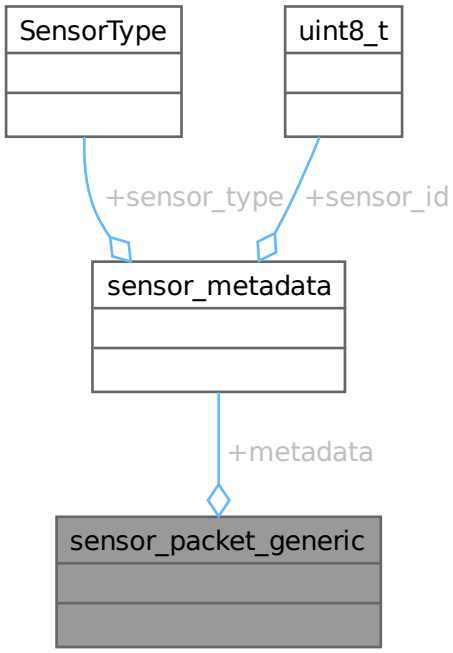
- include/packets.h

6.16 sensor_packet_generic Struct Reference

Structure for generic sensor packets.

```
#include <packets.h>
```

Collaboration diagram for sensor_packet_generic:



Public Attributes

- struct sensor_metadata metadata

6.16.1 Detailed Description

Structure for generic sensor packets.

This structure contains the type and ID of the sensor being addressed. This structure is used for generic sensor packets that do not require additional data. For example, it can be used for a simple button press event.

Definition at line 82 of file packets.h.

6.16.2 Member Data Documentation

6.16.2.1 metadata

```
struct sensor_metadata sensor_packet_generic::metadata
```

Definition at line 83 of file packets.h.

The documentation for this struct was generated from the following file:

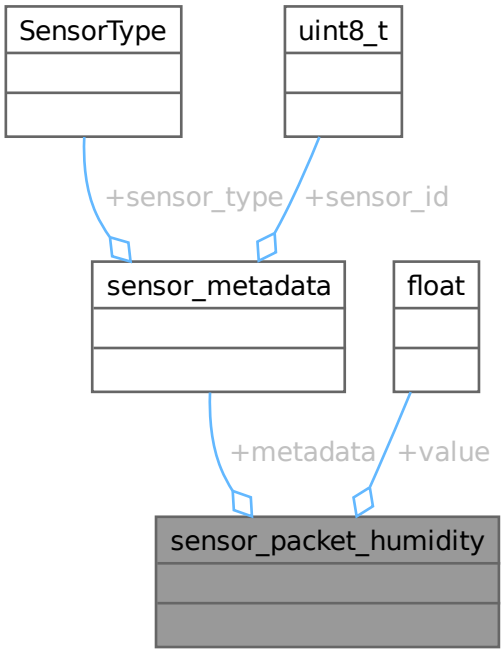
- include/packets.h

6.17 sensor_packet_humidity Struct Reference

Structure for humidity sensor packets.

```
#include <packets.h>
```

Collaboration diagram for sensor_packet_humidity:



Public Attributes

- struct sensor_metadata metadata
- float value

  Value of the sensor reading the humidity level represented in percentage.

6.17.1 Detailed Description

Structure for humidity sensor packets.

This structure contains the type, ID, and value of the humidity sensor reading.

Note

The humidity value is represented in percentage.

Definition at line 121 of file packets.h.

6.17.2 Member Data Documentation

6.17.2.1 metadata

```
struct sensor_metadata sensor_packet_humidity::metadata
```

Definition at line 122 of file packets.h.

6.17.2.2 value

```
float sensor_packet_humidity::value
```

Value of the sensor reading the humidity level represented in percentage.

Definition at line 124 of file packets.h.

The documentation for this struct was generated from the following file:

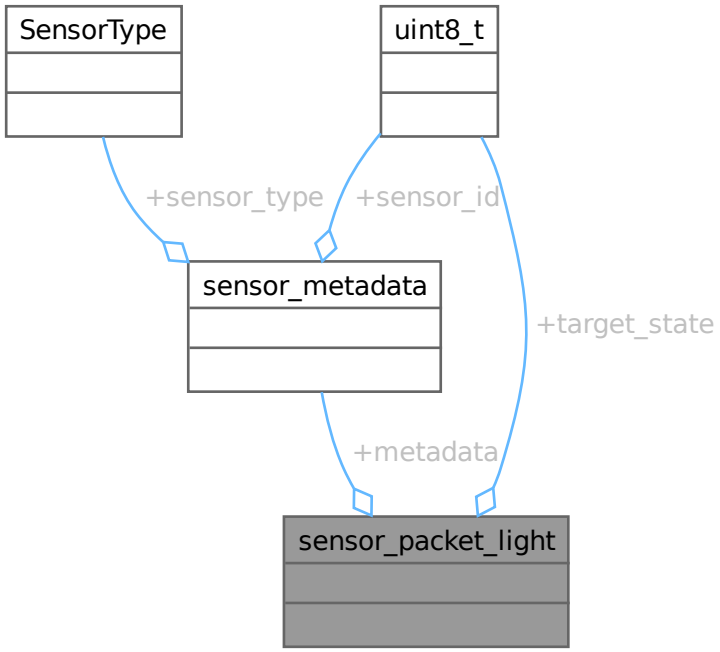
- include/packets.h

6.18 sensor_packet_light Struct Reference

Structure for light sensor packets.

`#include <packets.h>`

Collaboration diagram for sensor_packet_light:



Public Attributes

- struct sensor_metadata metadata
- uint8_t target_state

  Target state of the light (on 1/off 0) represented as a boolean value.

6.18.1 Detailed Description

Structure for light sensor packets.

This structure contains the type, ID, and target state of the light/led. This structure is used for light control packets sent to the light/led.

Definition at line 134 of file packets.h.

6.18.2 Member Data Documentation

6.18.2.1 metadata

```
struct sensor_metadata sensor_packet_light::metadata
```

Definition at line 135 of file packets.h.

6.18.2.2 target_state

```
uint8_t sensor_packet_light::target_state
```

Target state of the light (on 1/off 0) represented as a boolean value.

Definition at line 137 of file packets.h.

The documentation for this struct was generated from the following file:

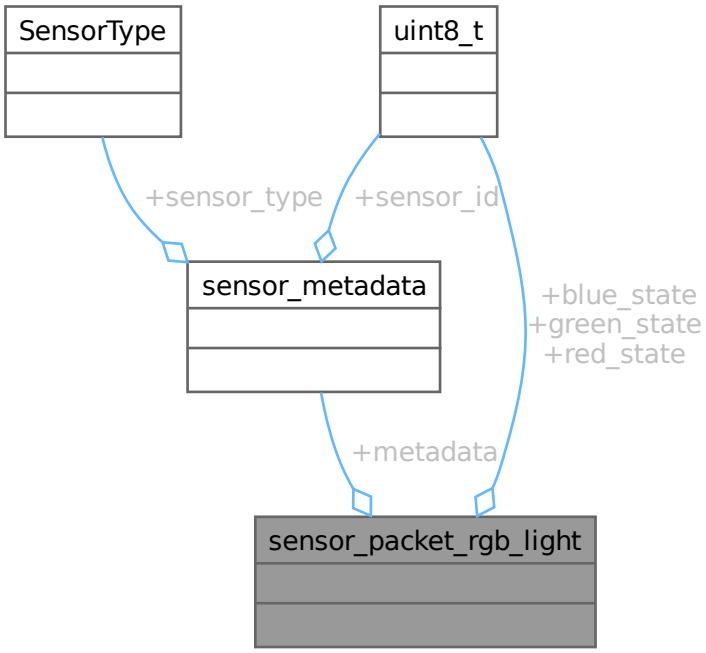
- include/packets.h

6.19 sensor_packet_rgb_light Struct Reference

Structure for RGB light sensor packets.

```
#include <packets.h>
```

Collaboration diagram for sensor_packet_rgb_light:



Public Attributes

- struct sensor_metadata metadata
- uint8_t red_state

  Target state of the red color (0-255) represented as an 8-bit integer.
- uint8_t green_state

  Target state of the green color (0-255) represented as an 8-bit integer.
- uint8_t blue_state

  Target state of the blue color (0-255) represented as an 8-bit integer.

6.19.1 Detailed Description

Structure for RGB light sensor packets.

This structure contains the type, ID, and target color of the RGB light. This structure is used for RGB light control packets sent to the RGB light.

Note

The RGB values are represented as 8-bit integers (0-255).

Definition at line 148 of file packets.h.

6.19.2 Member Data Documentation

6.19.2.1 blue_state

```
uint8_t sensor_packet_rgb_light::blue_state
```

Target state of the blue color (0-255) represented as an 8-bit integer.

Definition at line 155 of file packets.h.

6.19.2.2 green_state

```
uint8_t sensor_packet_rgb_light::green_state
```

Target state of the green color (0-255) represented as an 8-bit integer.

Definition at line 153 of file packets.h.

6.19.2.3 metadata

```
struct sensor_metadata sensor_packet_rgb_light::metadata
```

Definition at line 149 of file packets.h.

6.19.2.4 red_state

```
uint8_t sensor_packet_rgb_light::red_state
```

Target state of the red color (0-255) represented as an 8-bit integer.

Definition at line 151 of file packets.h.

The documentation for this struct was generated from the following file:

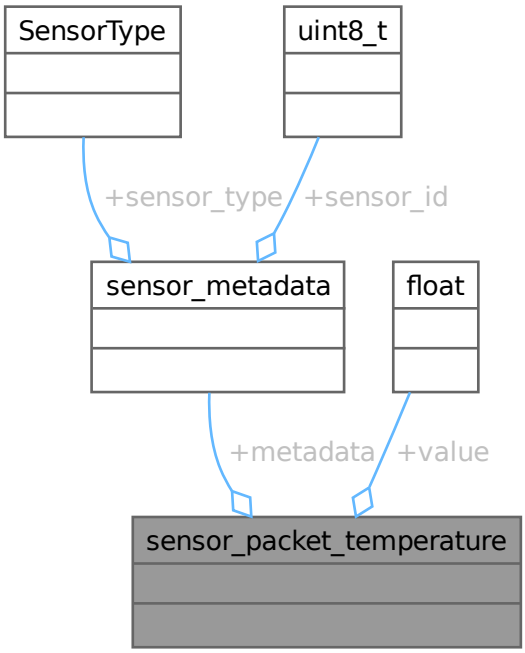
- include/packets.h

6.20 sensor_packet_temperature Struct Reference

Structure for temperature sensor packets.

```
#include <packets.h>
```

Collaboration diagram for sensor_packet_temperature:



Public Attributes

- struct sensor_metadata metadata
- float value

  Value of the sensor reading the temperature represented in Celcius.

6.20.1 Detailed Description

Structure for temperature sensor packets.

This structure contains the type, ID, and value of the temperature sensor reading.

Note

The temperature value is represented in Celsius.

Definition at line 95 of file packets.h.

6.20.2 Member Data Documentation

6.20.2.1 metadata

```
struct sensor_metadata sensor_packet_temperature::metadata
```

Definition at line 96 of file packets.h.

6.20.2.2 value

```
float sensor_packet_temperature::value
```

Value of the sensor reading the temperature represented in Celcius.

Definition at line 98 of file packets.h.

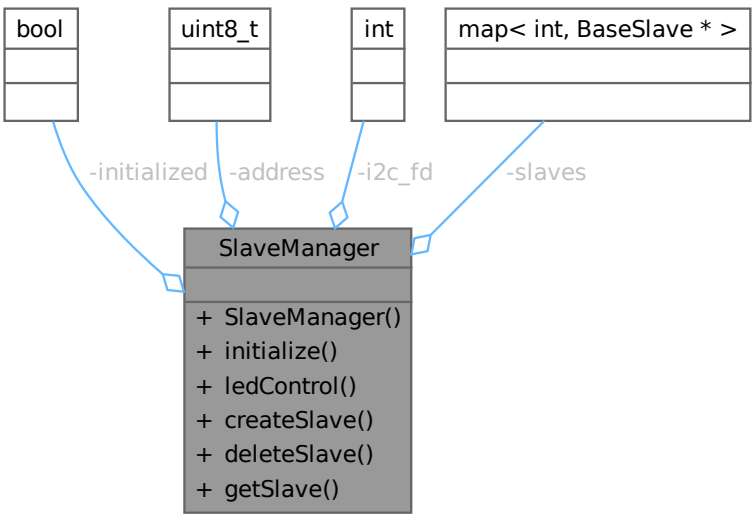
The documentation for this struct was generated from the following file:

- include/packets.h

6.21 SlaveManager Class Reference

```
#include <SlaveManager.h>
```

Collaboration diagram for SlaveManager:



Public Member Functions

- SlaveManager ()
- void initialize ()

  Setup the underlying I2C bus.
- void ledControl (uint8_t led_number, uint8_t led_state)
- void createSlave (SensorType type, int i2c_address)
- void deleteSlave (uint8_t id)

  Unmaps the slave from the internal mapping.
- BaseSlave * getSlave (int id)

  Get a slavedevice with a given ID.

Private Attributes

- bool initialized = false
- uint8_t address = 0x01
- int i2c_fd = -1
- std::map< int, BaseSlave * > slaves

6.21.1 Detailed Description

Definition at line 9 of file SlaveManager.h.

6.21.2 Constructor & Destructor Documentation

6.21.2.1 SlaveManager()

```
SlaveManager::SlaveManager ( )
```

Definition at line 15 of file SlaveManager.cpp.

6.21.3 Member Function Documentation

6.21.3.1 createSlave()

```
void SlaveManager::createSlave (
            SensorType type,
            int i2c_address )
```

Definition at line 30 of file SlaveManager.cpp.

6.21.3.2 deleteSlave()

```
void SlaveManager::deleteSlave (
            uint8_t id )
```

Unmaps the slave from the internal mapping.

Parameters

id	The ID of the slave device to unmap

Definition at line 47 of file SlaveManager.cpp.

6.21.3.3 getSlave()

```
BaseSlave * SlaveManager::getSlave (
            int id )
```

Get a slavedevice with a given ID.

Parameters

id	The ID of the slave device to retrieve

Definition at line 56 of file SlaveManager.cpp.

6.21.3.4 initialize()

```
void SlaveManager::initialize ( )
```

Setup the underlying I2C bus.

Exceptions

std__runtime_error	if the I2C setup fails

Definition at line 17 of file SlaveManager.cpp.

6.21.3.5 ledControl()

```
void SlaveManager::ledControl (
            uint8_t led_number,
            uint8_t led_state )
```

Definition at line 28 of file SlaveManager.cpp.

6.21.4 Member Data Documentation

6.21.4.1 address

```
uint8_t SlaveManager::address = 0x01  [private]
```

Definition at line 46 of file SlaveManager.h.

6.21.4.2 i2c_fd

```
int SlaveManager::i2c_fd = -1  [private]
```

Definition at line 47 of file SlaveManager.h.

6.21.4.3 initialized

```
bool SlaveManager::initialized = false  [private]
```

Definition at line 44 of file SlaveManager.h.

6.21.4.4 slaves

```
std::map<int, BaseSlave*> SlaveManager::slaves  [private]
```

Definition at line 49 of file SlaveManager.h.

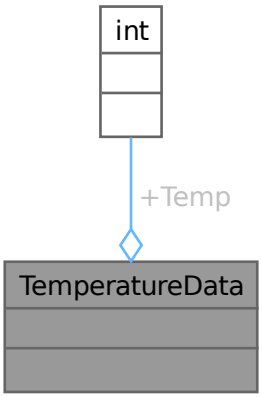
The documentation for this class was generated from the following files:

- include/SlaveManager.h
- src/SlaveManager.cpp

6.22 TemperatureData Struct Reference

```
#include <TemperatureSlave.h>
```

Collaboration diagram for TemperatureData:



Public Attributes

- int Temp

6.22.1 Detailed Description

Definition at line 6 of file TemperatureSlave.h.

6.22.2 Member Data Documentation

6.22.2.1 Temp

```
int TemperatureData::Temp
```

Definition at line 7 of file TemperatureSlave.h.

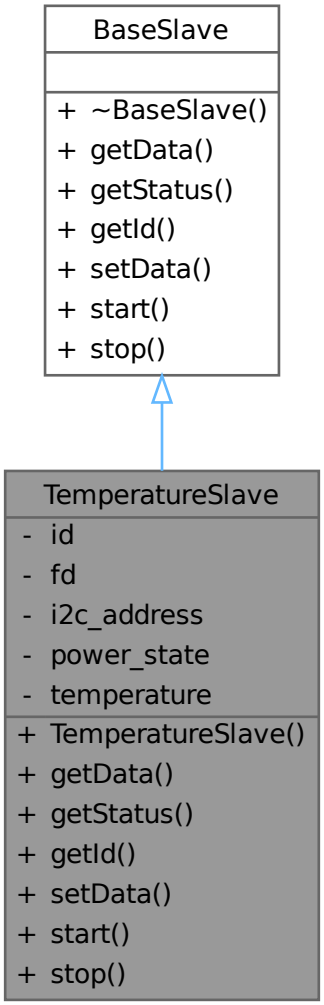
The documentation for this struct was generated from the following file:

- include/TemperatureSlave.h

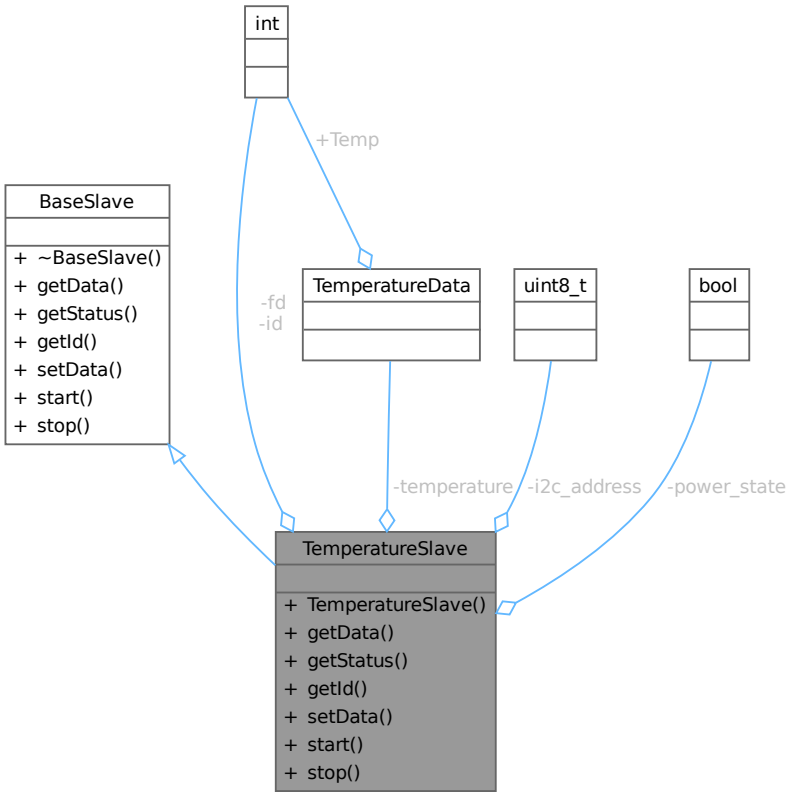
6.23 TemperatureSlave Class Reference

```
#include <TemperatureSlave.h>
```

Inheritance diagram for TemperatureSlave:



Collaboration diagram for TemperatureSlave:



Public Member Functions

- TemperatureSlave (uint8_t id, uint8_t i2c_address)
- void * getData ()
- bool getStatus ()
- int getId ()
- void setData (void *data)
- void start (int i2c_fd)
- void stop ()

Public Member Functions inherited from BaseSlave

- virtual ~BaseSlave ()=default

Private Attributes

- int id
- int fd
- uint8_t i2c_address
- bool power_state
- TemperatureData temperature

6.23.1 Detailed Description

Definition at line 10 of file TemperatureSlave.h.

6.23.2 Constructor & Destructor Documentation

6.23.2.1 TemperatureSlave()

```
TemperatureSlave::TemperatureSlave (
            uint8_t id,
            uint8_t i2c_address )
```

Definition at line 3 of file TemperatureSlave.cpp.

6.23.3 Member Function Documentation

6.23.3.1 getData()

```
void * TemperatureSlave::getData ( )  [virtual]
```

Implements BaseSlave.

Definition at line 6 of file TemperatureSlave.cpp.

6.23.3.2 getId()

```
int TemperatureSlave::getId ( )  [virtual]
```

Implements BaseSlave.

Definition at line 23 of file TemperatureSlave.cpp.

6.23.3.3 getStatus()

```
bool TemperatureSlave::getStatus ( )  [virtual]
```

Implements BaseSlave.

Definition at line 14 of file TemperatureSlave.cpp.

6.23.3.4 setData()

```
void TemperatureSlave::setData (
            void * data )  [virtual]
```

Implements BaseSlave.

Definition at line 25 of file TemperatureSlave.cpp.

6.23.3.5 start()

```
void TemperatureSlave::start (
            int i2c_fd )  [virtual]
```

Implements BaseSlave.

Definition at line 32 of file TemperatureSlave.cpp.

6.23.3.6 stop()

```
void TemperatureSlave::stop ( )  [virtual]
```

Implements BaseSlave.

Definition at line 34 of file TemperatureSlave.cpp.

6.23.4 Member Data Documentation

6.23.4.1 fd

```
int TemperatureSlave::fd  [private]
```

Definition at line 22 of file TemperatureSlave.h.

6.23.4.2 i2c_address

```
uint8_t TemperatureSlave::i2c_address  [private]
```

Definition at line 23 of file TemperatureSlave.h.

6.23.4.3 id

```
int TemperatureSlave::id  [private]
```

Definition at line 21 of file TemperatureSlave.h.

6.23.4.4 power_state

```
bool TemperatureSlave::power_state  [private]
```

Definition at line 25 of file TemperatureSlave.h.

6.23.4.5 temperature

```
TemperatureData TemperatureSlave::temperature  [private]
```

Definition at line 26 of file TemperatureSlave.h.

The documentation for this class was generated from the following files:

- include/TemperatureSlave.h
- src/TemperatureSlave.cpp

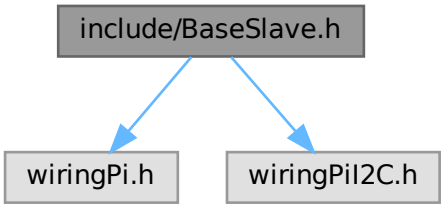
Chapter 7

File Documentation

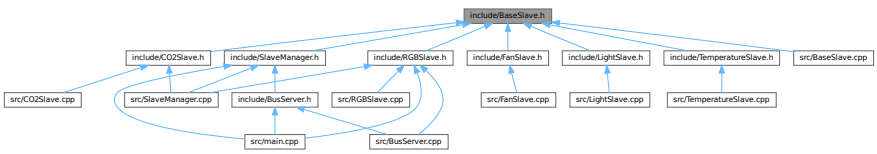
7.1 include/BaseSlave.h File Reference

```
#include <wiringPi.h>
#include <wiringPiI2C.h>
```

Include dependency graph for BaseSlave.h:



This graph shows which files directly or indirectly include this file:



Classes

- class BaseSlave

7.2 BaseSlave.h

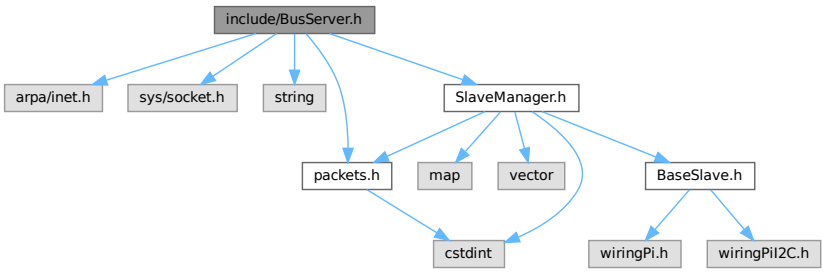
Go to the documentation of this file.

```
00001 #pragma once
00002
00003 #include <wiringPi.h>
00004 #include <wiringPiI2C.h>
00005
00006 class BaseSlave {
00007    public:
00008     virtual ~BaseSlave() = default;
00009     virtual const void* getData() = 0;
00010     virtual bool getStatus() = 0;
00011     virtual int getId() = 0;
00012     virtual void setData(void* data) = 0;
00013     virtual void start(int i2c_fd) = 0;
00014     virtual void stop() = 0;
00015 };
```

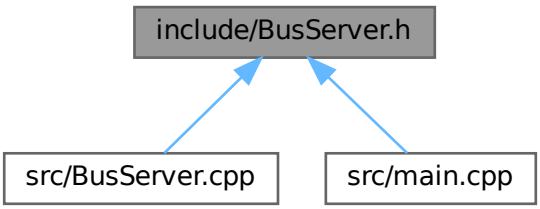
7.3 include/BusServer.h File Reference

```
#include <arpa/inet.h>
#include <sys/socket.h>
#include <string>
#include "SlaveManager.h"
#include "packets.h"
```

Include dependency graph for BusServer.h:



This graph shows which files directly or indirectly include this file:



Classes

- class BusServer

Macros

- #define BUFFER_SIZE 1024

7.3.1 Macro Definition Documentation

7.3.1.1 BUFFER_SIZE

```
#define BUFFER_SIZE 1024
```

Definition at line 10 of file BusServer.h.

7.4 BusServer.h

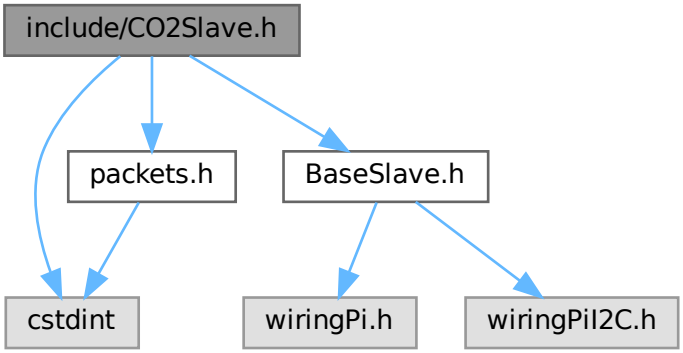
Go to the documentation of this file.

```
00001 #pragma once
00002 #include <arpa/inet.h>
00003 #include <sys/socket.h>
00004
00005 #include <string>
00006
00007 #include "SlaveManager.h"
00008 #include "packets.h"
00009
00010 #define BUFFER_SIZE 1024
00011
00012 class BusServer {
00013    public:
00014     BusServer() : listening_fd(-1) {}
00015
00025     void setup(std::string ip, int port);
00026
00031     void listen();
00032
00041     void send(struct sensor_packet* pkt, int fd);
00042
00048     void start();
00049
00053     SlaveManager& getSlaveManager();
00054
00055    private:
00056     int listening_fd;
00057
00058     struct sockaddr_in listening_address;
00059     bool wemos_bridge_connected = false;
00060     char buffer[BUFFER_SIZE];
00061
00062     SlaveManager slave_manager;
00063 };
```

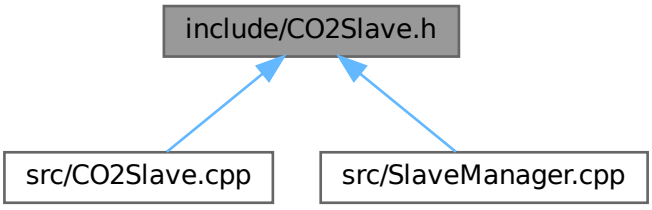
7.5 include/CO2Slave.h File Reference

```
#include <cstdint>
#include "BaseSlave.h"
#include "packets.h"
```

Include dependency graph for CO2Slave.h:



This graph shows which files directly or indirectly include this file:



Classes

- class CO2Slave

7.6 CO2Slave.h

Go to the documentation of this file.

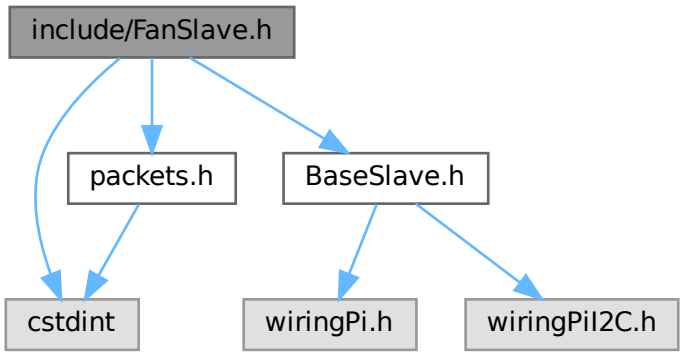
```
00001 #pragma once
00002 #include <cstdint>
00003
00004 #include "BaseSlave.h"
```

```
00005 #include "packets.h"
00006
00007 class CO2Slave : public BaseSlave {
00008    public:
00009     CO2Slave(uint8_t id, uint8_t i2c_address);
00010
00011     const void* getData() override;
00012     bool getStatus() override;
00013     int getId() override;
00014     void setData(void* data) override;
00015     void start(int i2c_fd) override;
00016     void stop() override;
00017
00018    private:
00019     int id;
00020     int fd;
00021     uint8_t i2c_address;
00022     uint16_t command;
00023
00024     bool power_state = true;
00025     struct sensor_packet state_packet;
00026 };
```

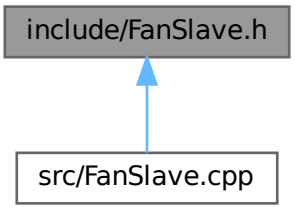
7.7 include/FanSlave.h File Reference

```
#include <cstdint>
#include "BaseSlave.h"
#include "packets.h"
```

Include dependency graph for FanSlave.h:



This graph shows which files directly or indirectly include this file:



Classes

- class FanSlave

Typedefs

- typedef uint8_t FanData

7.7.1 Typedef Documentation

7.7.1.1 FanData

```
typedef uint8_t FanData
```

Definition at line 8 of file FanSlave.h.

7.8 FanSlave.h

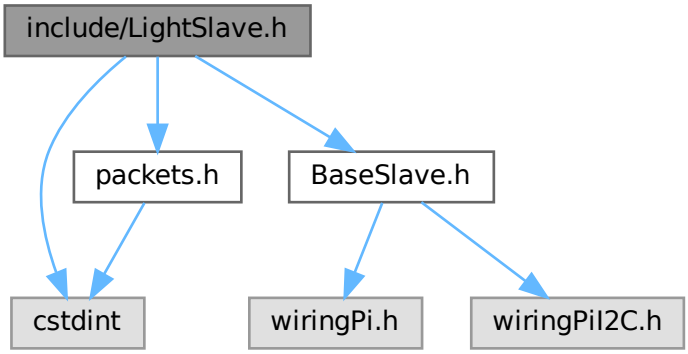
Go to the documentation of this file.

```
00001 #pragma once
00002
00003 #include <cstdint>
00004
00005 #include "BaseSlave.h"
00006 #include "packets.h"
00007
00008 typedef uint8_t FanData;
00009
00010 class FanSlave : public BaseSlave {
00011    public:
00012     FanSlave(uint8_t id, uint8_t i2c_address);
00013     const void* getData();
00014     bool getStatus();
00015     int getId();
00016     void setData(void* data);
00017     void start(int i2c_fd);
00018     void stop();
00019
00020    private:
00021     int id;
00022     int fd;
00023     uint8_t i2c_address;
00024
00025     FanData current_speed;
00026
00027     struct sensor_packet state_packet;
00028 };
```

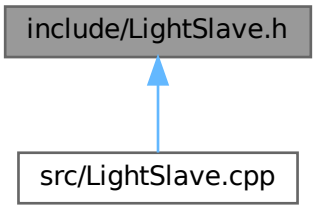
7.9 include/LightSlave.h File Reference

```
#include <cstdint>
#include "BaseSlave.h"
#include "packets.h"
```

Include dependency graph for LightSlave.h:



This graph shows which files directly or indirectly include this file:



Classes

- class LightSlave

Typedefs

- typedef uint8_t LightData

7.9.1 Typedef Documentation

7.9.1.1 LightData

```
typedef uint8_t LightData
```

Definition at line 7 of file LightSlave.h.

7.10 LightSlave.h

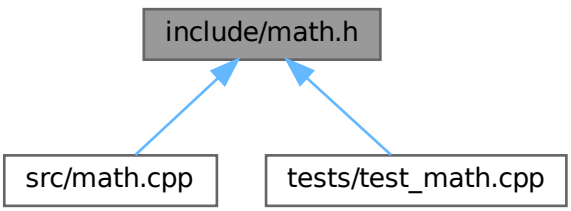
Go to the documentation of this file.

```
00001 #pragma once
00002 #include <cstdint>
00003
00004 #include "BaseSlave.h"
00005 #include "packets.h"
00006
00007 typedef uint8_t LightData;
00008
00009 class LightSlave : public BaseSlave {
00010    public:
00011     LightSlave(uint8_t id, uint8_t i2c_address);
00012     const void* getData();
00013     bool getStatus();
00014     int getId();
00015     void setData(void* data);
00016     void start(int i2c_fd);
00017     void stop();
00018
00019    private:
00020     int id;
00021     int fd;
00022     uint8_t i2c_address;
00023
00024     LightData color_state;
00025
00026     struct sensor_packet state_packet;
00027 };
```

7.11 include/math.h File Reference

Header file for math.cpp.

This graph shows which files directly or indirectly include this file:



Functions

- int add (int a, int b)

  Adds two integers.
- int subtract (int a, int b)

  Subtracts two integers.

7.11.1 Detailed Description

Header file for math.cpp.

This file contains declarations for basic math operations.

Author

Daan Breur

Definition in file math.h.

7.11.2 Function Documentation

7.11.2.1 add()

```
int add (
            int a,
            int b )
```

Adds two integers.

Parameters

a	First integer.
b	Second integer.

Returns

The sum of a and b.

This function takes two integers as input and returns their sum.

Definition at line 16 of file math.cpp.

7.11.2.2 subtract()

```
int subtract (
            int a,
            int b )
```

Subtracts two integers.

Parameters

a	First integer.
b	Second integer.

Returns

The difference of a and b.

This function takes two integers as input and returns the result of subtracting b from a.

Definition at line 26 of file math.cpp.

7.12 math.h

Go to the documentation of this file.

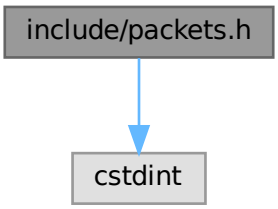
```
00001
00008 #ifndef MATH_H
00009 #define MATH_H
00010
00011 int add(int a, int b);
00012 int subtract(int a, int b);
00013
00014 #endif
```

7.13 include/packets.h File Reference

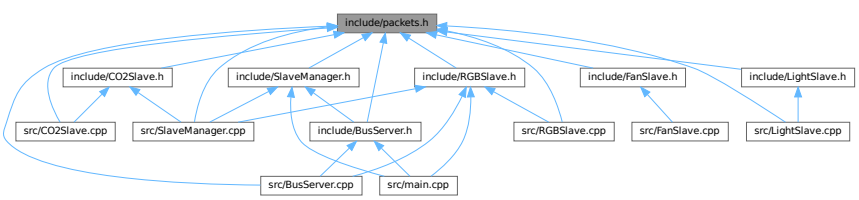
Header file for packets.h.

```
#include <cstdint>
```

Include dependency graph for packets.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct sensor_header

  Header structure for sensor packets.
- struct sensor_metadata

  Structure for sensor metadata, which is always included in any packet.
- struct sensor_heartbeat

  Structure for heartbeat packets.
- struct sensor_packet_generic

  Structure for generic sensor packets.
- struct sensor_packet_temperature

  Structure for temperature sensor packets.
- struct sensor_packet_co2

  Structure for CO2 sensor packets.
- struct sensor_packet_humidity

  Structure for humidity sensor packets.
- struct sensor_packet_light

  Structure for light sensor packets.
- struct sensor_packet_rgb_light

  Structure for RGB light sensor packets.
- struct sensor_packet_fan

  Structure for packets intended to control the fan speed.
- struct sensor_packet

  Union structure for the entire sensor packet.
- union sensor_packet::sensor_data

- union sensor_data

Enumerations

- enum class SensorType : uint8_t {
  NOOP = 0 , BUTTON = 1 , TEMPERATURE = 2 , CO2 = 3 ,
  HUMIDITY = 4 , PRESSURE = 5 , LIGHT = 6 , MOTION = 7 ,
  RGB_LIGHT = 8 , FAN = 10 }
- enum class PacketType : uint8_t {
  DATA = 0 , HEARTBEAT = 1 , DASHBOARD_POST = 2 , DASHBOARD_GET = 3 ,
  DASHBOARD_RESPONSE = 4 }

Functions

- struct sensor_header __attribute__ ((packed))

Variables

- uint8_t length

  Length of the packet excluding the header.
- PacketType ptype

  Type of the packet as PacketType (DATA, HEARTBEAT, etc.).
- SensorType sensor_type

  Type of the sensor being addressed as SensorType (one byte)
- uint8_t sensor_id

  ID of the sensor being addressed.
- struct sensor_metadata metadata
- float value

  Value of the sensor reading the temperature represented in Celcius.
- uint8_t target_state

  Target state of the light (on 1/off 0) represented as a boolean value.
- uint8_t red_state

  Target state of the red color (0-255) represented as an 8-bit integer.
- uint8_t green_state

  Target state of the green color (0-255) represented as an 8-bit integer.
- uint8_t blue_state

  Target state of the blue color (0-255) represented as an 8-bit integer.
- uint8_t speed

  Target speed of the fan (0 -> 255 = 0 -> 100%)
- struct sensor_header header

  Header of the packet containing length and type information.
- union sensor_data data

7.13.1 Detailed Description

Header file for packets.h.

This files origin is from the Wemos project

Warning

THIS FILE MUST BE KEPT IN SYNC IN OTHER PROJECTS

Author

Daan Breur

Erynn Scholtes

Definition in file packets.h.

7.13.2 Enumeration Type Documentation

7.13.2.1 PacketType

```
enum class PacketType : uint8_t [strong]
```

Enumerator

DATA	
HEARTBEAT	
DASHBOARD_POST	
DASHBOARD_GET	
DASHBOARD_RESPONSE	

Definition at line 28 of file packets.h.

7.13.2.2 SensorType

```
enum class SensorType :  uint8_t  [strong]
```

Enumerator

NOOP	
BUTTON	
TEMPERATURE	
CO2	
HUMIDITY	
PRESSURE	
LIGHT	
MOTION	
RGB_LIGHT	
FAN	

Definition at line 15 of file packets.h.

7.13.3 Function Documentation

7.13.3.1 __attribute__()

```
struct sensor_header __attribute__ (
            (packed)  )
```

7.13.4 Variable Documentation

7.13.4.1 blue_state

```
uint8_t blue_state
```

Target state of the blue color (0-255) represented as an 8-bit integer.

Definition at line 6 of file packets.h.

7.13.4.2 data

```
union sensor_data data
```

7.13.4.3 green_state

```
uint8_t green_state
```

Target state of the green color (0-255) represented as an 8-bit integer.

Definition at line 4 of file packets.h.

7.13.4.4 header

```
struct sensor_header header
```

Header of the packet containing length and type information.

Definition at line 1 of file packets.h.

7.13.4.5 length

```
uint8_t length
```

Length of the packet excluding the header.

Definition at line 1 of file packets.h.

7.13.4.6 metadata

```
struct sensor_metadata metadata
```

Definition at line 0 of file packets.h.

7.13.4.7 ptype

```
PacketType ptype
```

Type of the packet as PacketType (DATA, HEARTBEAT, etc.).

Definition at line 3 of file packets.h.

7.13.4.8 red_state

```
uint8_t red_state
```

Target state of the red color (0-255) represented as an 8-bit integer.

Definition at line 2 of file packets.h.

7.13.4.9 sensor_id

```
uint8_t sensor_id
```

ID of the sensor being addressed.

Definition at line 3 of file packets.h.

7.13.4.10 sensor_type

```
SensorType sensor_type
```

Type of the sensor being addressed as SensorType (one byte)

Definition at line 1 of file packets.h.

7.13.4.11 speed

```
uint8_t speed
```

Target speed of the fan (0 -> 255 = 0 -> 100%)

Definition at line 2 of file packets.h.

7.13.4.12 target_state

```
uint8_t target_state
```

Target state of the light (on 1/off 0) represented as a boolean value.

Definition at line 2 of file packets.h.

7.13.4.13 value

```
float value
```

Value of the sensor reading the temperature represented in Celcius.

Value of the sensor reading the humidity level represented in percentage.

Value of the sensor reading the CO2 level represented in ppm.

Definition at line 2 of file packets.h.

7.14 packets.h

Go to the documentation of this file.

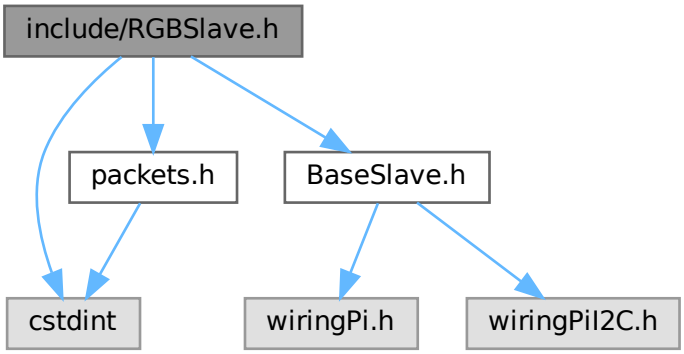
```
00001
00010 #ifndef PACKETS_H
00011 #define PACKETS_H
00012
00013 #include <cstdint>
00014
00015 enum class SensorType : uint8_t {
00016     NOOP = 0,
00017     BUTTON = 1,
00018     TEMPERATURE = 2,
00019     CO2 = 3,
00020     HUMIDITY = 4,
00021     PRESSURE = 5,
00022     LIGHT = 6,
00023     MOTION = 7,
00024     RGB_LIGHT = 8,
00025     FAN = 10,
00026 };
00027
00028 enum class PacketType : uint8_t {
00029     DATA = 0,
00030     HEARTBEAT = 1,
00031     DASHBOARD_POST = 2,
00032     DASHBOARD_GET = 3,
00033     DASHBOARD_RESPONSE = 4
00034 };
00035
00041 struct sensor_header {
00043     uint8_t length;
00045     PacketType ptype;
00046 } __attribute__((packed));
00047
00053 struct sensor_metadata {
00055     SensorType sensor_type;
00057     uint8_t sensor_id;
00058 } __attribute__((packed));
00059
00060 // Specific packet structures (ensure alignment/packing matches expected format)
00061
00070 struct sensor_heartbeat {
00071     struct sensor_metadata metadata;
00072 } __attribute__((packed));
00073
00082 struct sensor_packet_generic {
00083     struct sensor_metadata metadata;
00084     // /** @brief Whether the sensor did or did not trigger */
00085     // bool value;
00086 } __attribute__((packed));
00087
00095 struct sensor_packet_temperature {
00096     struct sensor_metadata metadata;
00098     float value;
00099 } __attribute__((packed));
00100
00108 struct sensor_packet_co2 {
00109     struct sensor_metadata metadata;
00111     uint16_t value;
00112 } __attribute__((packed));
00113
00121 struct sensor_packet_humidity {
00122     struct sensor_metadata metadata;
00124     float value;
00125 } __attribute__((packed));
00126
00134 struct sensor_packet_light {
00135     struct sensor_metadata metadata;
00137     uint8_t target_state;
00138 } __attribute__((packed));
00139
00148 struct sensor_packet_rgb_light {
00149     struct sensor_metadata metadata;
00151     uint8_t red_state;
00153     uint8_t green_state;
00155     uint8_t blue_state;
00156 } __attribute__((packed));
00157
00164 struct sensor_packet_fan {
00165     struct sensor_metadata metadata;
00167     uint8_t speed;
00168 } __attribute__((packed));
00169 // --- End Structures ---
00170
```

```
00235 struct sensor_packet {
00237     struct sensor_header header;
00238
00240     union sensor_data {
00241         struct sensor_heartbeat heartbeat;
00242         struct sensor_packet_generic generic;
00243         struct sensor_packet_temperature temperature;
00244         struct sensor_packet_co2 co2;
00245         struct sensor_packet_humidity humidity;
00246         struct sensor_packet_light light;
00247         struct sensor_packet_rgb_light rgb_light;
00248     } data;
00249 } __attribute__((packed));
00250
00251 #endif  // PACKETS_H
```

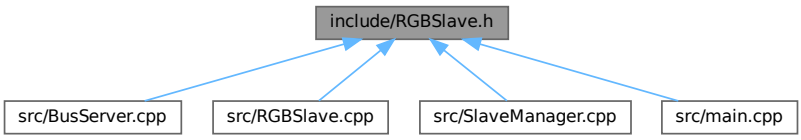
7.15 include/RGBSlave.h File Reference

```
#include <cstdint>
#include "BaseSlave.h"
#include "packets.h"
```

Include dependency graph for RGBSlave.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct RGBData
- class RGBSlave

Functions

- struct RGBData __attribute__ ((packed))

Variables

- uint8_t R
- uint8_t G
- uint8_t B
- RGBSlave __attribute__

7.15.1 Function Documentation

7.15.1.1 __attribute__()

```
struct RGBData __attribute__ (
            (packed)  )
```

7.15.2 Variable Documentation

7.15.2.1 __attribute__

```
struct sensor_packet __attribute__
```

7.15.2.2 B

```
uint8_t B
```

Definition at line 0 of file RGBSlave.h.

7.15.2.3 G

```
uint8_t G
```

Definition at line 0 of file RGBSlave.h.

7.15.2.4 R

```
uint8_t R
```

Definition at line 0 of file RGBSlave.h.

7.16 RGBSlave.h

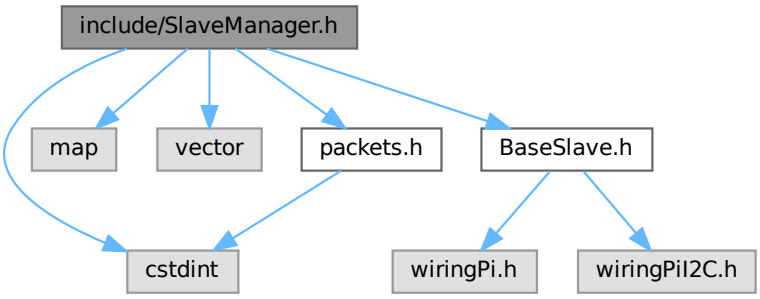
Go to the documentation of this file.

```
00001 #pragma once
00002 #include <cstdint>
00003
00004 #include "BaseSlave.h"
00005 #include "packets.h"
00006
00007 struct RGBData {
00008     uint8_t R, G, B;
00009 } __attribute__((packed));
00010
00011 class RGBSlave : public BaseSlave {
00012    public:
00013     RGBSlave(uint8_t id, uint8_t i2c_address);
00014     const void* getData();
00015     bool getStatus();
00016     int getId();
00017     void setData(void* data);
00018     void start(int i2c_fd);
00019     void stop();
00020
00021    private:
00022     int id;
00023     int fd;
00024     uint8_t i2c_address;
00025
00026     RGBData color_state;
00027
00028     struct sensor_packet state_packet;
00029 };
```

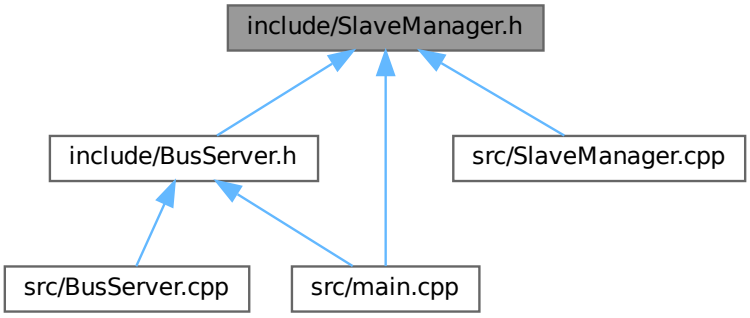
7.17 include/SlaveManager.h File Reference

```
#include <cstdint>
#include <map>
#include <vector>
#include "BaseSlave.h"
#include "packets.h"
```

Include dependency graph for SlaveManager.h:



This graph shows which files directly or indirectly include this file:



Classes

- class SlaveManager

7.18 SlaveManager.h

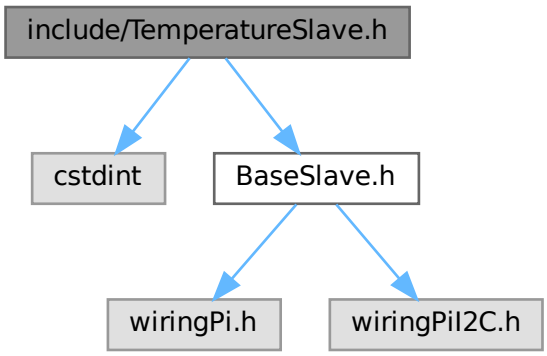
Go to the documentation of this file.

```
00001 #pragma once
00002 #include <cstdint>
00003 #include <map>
00004 #include <vector>
00005
00006 #include "BaseSlave.h"
00007 #include "packets.h"
00008
00009 class SlaveManager {
00010    public:
00011     SlaveManager();
00012
00017     void initialize();
00018
00019     void ledControl(uint8_t led_number, uint8_t led_state);
00020
00021     /*
00022      * @brief Creates a new slave device into the internal mapping
00023      * @details Assigns a pre-known SensorType and ID to an I2C device, and then opens the I2C
00024      * connection to it
00025      * @param type The SensorType of the sensor, as defined in packets.h
00026      * @param id The ID of the slave device
00027      * @param i2c_address The I2C address of the slave device
00028      */
00029     void createSlave(SensorType type, int i2c_address);
00030
00035     void deleteSlave(uint8_t id);
00036
00041     BaseSlave* getSlave(int id);
00042
00043    private:
00044     bool initialized = false;
00045
00046     uint8_t address = 0x01;
00047     int i2c_fd = -1;
00048
00049     std::map<int, BaseSlave*> slaves;  // maps ID to BaseSlave ptr
00050 };
```

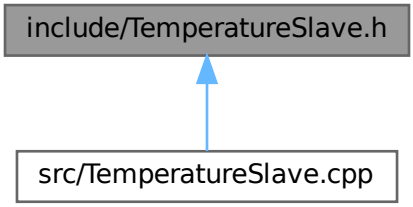
7.19 include/TemperatureSlave.h File Reference

```
#include <cstdint>
#include "BaseSlave.h"
```

Include dependency graph for TemperatureSlave.h:



This graph shows which files directly or indirectly include this file:



Classes

- struct TemperatureData
- class TemperatureSlave

Functions

- struct TemperatureData __attribute__ ((packed))

Variables

- int Temp
- TemperatureSlave __attribute__

7.19.1 Function Documentation

7.19.1.1 __attribute__()

```
struct TemperatureData __attribute__ (
             (packed)  )
```

7.19.2 Variable Documentation

7.19.2.1 __attribute__

```
TemperatureSlave __attribute__
```

7.19.2.2 Temp

```
int Temp
```

Definition at line 0 of file TemperatureSlave.h.

7.20 TemperatureSlave.h

Go to the documentation of this file.

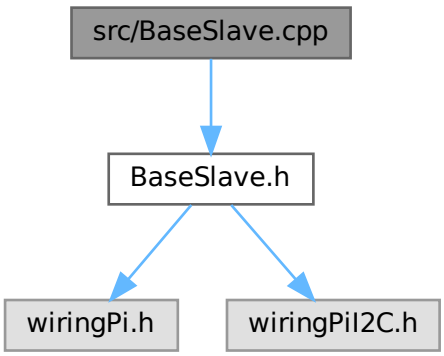
```
00001 #pragma once
00002 #include <cstdint>
00003
00004 #include "BaseSlave.h"
00005
00006 struct TemperatureData {
00007     int Temp;
00008 } __attribute__((packed));
00009
00010 class TemperatureSlave : public BaseSlave {
00011    public:
00012     TemperatureSlave(uint8_t id, uint8_t i2c_address);
00013     void* getData();
00014     bool getStatus();
00015     int getId();
00016     void setData(void* data);
00017     void start(int i2c_fd);
00018     void stop();
00019
00020    private:
00021     int id;
00022     int fd;
00023     uint8_t i2c_address;
00024
00025     bool power_state;
00026     TemperatureData temperature;
00027 };
```

7.21 README.md File Reference

7.22 src/BaseSlave.cpp File Reference

```
#include "BaseSlave.h"
```

Include dependency graph for BaseSlave.cpp:



7.23 BaseSlave.cpp

Go to the documentation of this file.

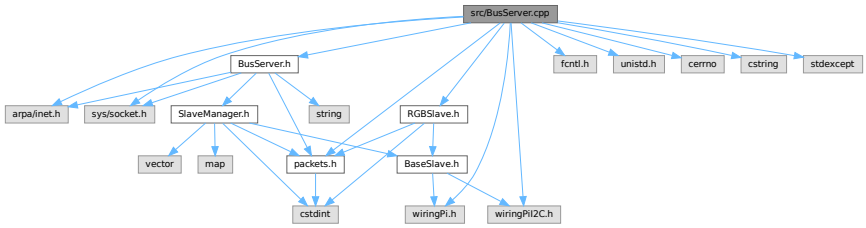
```
00001 #include "BaseSlave.h"
```

7.24 src/BusServer.cpp File Reference

```
#include "BusServer.h"
#include <arpa/inet.h>
#include <fcntl.h>
#include <sys/socket.h>
#include <unistd.h>
#include <cerrno>
#include <cstring>
#include <stdexcept>
#include "RGBSlave.h"
#include "packets.h"
#include "wiringPi.h"
```

```
#include "wiringPiI2C.h"
```

Include dependency graph for BusServer.cpp:



Macros

- #define RGB_SLAVE_ADDRESS 0x69

7.24.1 Macro Definition Documentation

7.24.1.1 RGB_SLAVE_ADDRESS

```
#define RGB_SLAVE_ADDRESS 0x69
```

Definition at line 17 of file BusServer.cpp.

7.25 BusServer.cpp

Go to the documentation of this file.

```
00001 #include "BusServer.h"
00002
00003 #include <arpa/inet.h>
00004 #include <fcntl.h>
00005 #include <sys/socket.h>
00006 #include <unistd.h>
00007
00008 #include <cerrno>
00009 #include <cstring>
00010 #include <stdexcept>
00011
00012 #include "RGBSlave.h"
00013 #include "packets.h"
00014 #include "wiringPi.h"
00015 #include "wiringPiI2C.h"
00016
00017 #define RGB_SLAVE_ADDRESS 0x69
00018
00019 void BusServer::setup(std::string ip, int port) {
00020     // slave_manager.initialize();
00021
00022     memset(&listening_address, 0, sizeof(listening_address));
00023
00024     if (port <= 0 || port > 65535) {
00025         throw std::invalid_argument("invalid port passed");
00026     }
00027
00028     if (0 == inet_aton(ip.c_str(), &listening_address.sin_addr)) {
00029         perror("inet_aton()");
00030         throw std::invalid_argument("invalid IP address passed");
00031     }
00032
00033     listening_address.sin_family = AF_INET;
00034     listening_address.sin_port = htons(port);
```

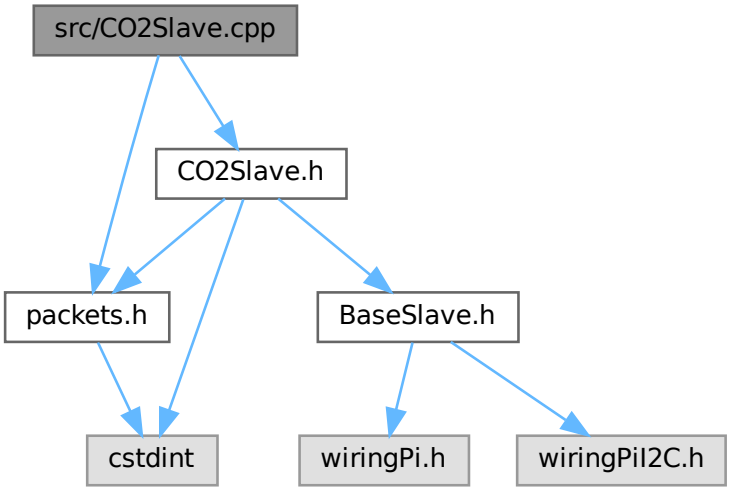
```
00035
00036     listening_fd = socket(AF_INET, SOCK_STREAM | SOCK_NONBLOCK, 0);
00037     if (-1 == listening_fd) {
00038         perror("socket()");
00039         throw std::runtime_error("failed to create socket");
00040     }
00041
00042     {
00043         int one = 1;
00044         if (-1 ==
00045             setsockopt(listening_fd, SOL_SOCKET, SO_REUSEADDR | SO_REUSEPORT, &one, sizeof(one))) {
00046             perror("setsockopt()");
00047             throw std::runtime_error("failed to set socket options");
00048         }
00049     }
00050
00051     socklen_t len = sizeof(listening_address);
00052
00053     if (-1 == bind(listening_fd, (struct sockaddr*)&listening_address, len)) {
00054         perror("bind()");
00055         throw std::runtime_error("failed to bind socket");
00056     }
00057 }
00058
00059 void BusServer::listen() {
00060     if (-1 == ::listen(listening_fd, 8)) {
00061         perror("listen()");
00062         throw std::runtime_error("failed to listen on socket");
00063     }
00064 }
00065
00066 void BusServer::send(struct sensor_packet* packet_ptr, int fd) {
00067     if (!wemos_bridge_connected) {
00068         throw std::runtime_error("no longer connected to socket");
00069     }
00070     if (-1 == ::send(fd, packet_ptr, sizeof(struct sensor_header) + packet_ptr->header.length, 0)) {
00071         perror("send()");
00072         throw std::runtime_error("failed to send on socket");
00073     }
00074 }
00075
00076 void BusServer::start() {
00077     // start off by mapping pre-known addresses to ID's
00078     listen();
00079
00080     int the_fd = -1;
00081
00082     while (true) {
00083         char buffer[32] = {0};
00084         bool net_request = false;
00085
00086         int new_fd = -1;
00087
00088         {
00089             struct sockaddr_in client_address = {0};
00090             socklen_t client_address_length = sizeof(client_address);
00091
00092             new_fd = accept4(listening_fd, (struct sockaddr*)&client_address,
00093                              &client_address_length, SOCK_NONBLOCK);
00094
00095             if (-1 == new_fd) {
00096                 int err = errno;
00097                 switch (err) {
00098                     case EWOULDBLOCK:
00099                         // no client has tried to connect; no big deal
00100                         break;
00101
00102                     default:
00103                         perror("accept4()");
00104                         throw std::runtime_error("accept4() failed");
00105                         break;
00106                 }
00107             } else {
00108                 the_fd = new_fd;
00109                 // printf("new fd: %d\nthe fd: %d\n", new_fd, the_fd);
00110             }
00111
00112             if (the_fd != -1) {
00113                 int err;
00114                 int recvd = recv(the_fd, buffer, sizeof(buffer), SOCK_NONBLOCK);
00115                 if (0 == recvd) {
00116                     close(the_fd);
00117                     the_fd = -1;
00118                     continue;
00119                 }
00120
00121                 if (-1 == recvd) err = errno;
```

```
00122                 // printf("%d\n", err);
00123                 // perror("recv balls");
00124                 if (-1 == recvd && err != EAGAIN) {
00125                     perror("recv()");
00126                     usleep(100000);
00127                     throw std::runtime_error("reading from client socket failed for some reason");
00128                 } else if (errno != EAGAIN) {
00129                     // printf("%d\n", recvd);
00130                 } else {
00131                     // printf("%d\n", recvd);
00132                     if (recvd > 0) net_request = true;
00133 
00134                     // perror("recv()[1]");
00135                     usleep(100000);
00136                 }
00137             }
00138         }
00139 
00140         if (net_request) {
00141             struct sensor_packet* pkt_ptr = (struct sensor_packet*)buffer;
00142             BaseSlave* the_slave = slave_manager.getSlave(pkt_ptr->data.generic.metadata.sensor_id);
00143             SensorType the_sensor_type = pkt_ptr->data.generic.metadata.sensor_type;
00144             uint8_t values[8] = {0};
00145 
00146             switch (pkt_ptr->header.ptype) {
00147                 case PacketType::DASHBOARD_POST:
00148                     switch (the_sensor_type) {
00149                         case SensorType::LIGHT:
00150                             values[0] = pkt_ptr->data.light.target_state;
00151 
00152                             the_slave->setData(values);
00153                             break;
00154 
00155                         case SensorType::RGB_LIGHT:
00156                             struct RGBData rgb_data = {.R = pkt_ptr->data.rgb_light.red_state,
00157                                                        .G = pkt_ptr->data.rgb_light.green_state,
00158                                                        .B = pkt_ptr->data.rgb_light.blue_state};
00159 
00160                             the_slave->setData(&rgb_data);
00161 
00162                             break;
00163                     }
00164                     break;
00165 
00166                 case PacketType::DASHBOARD_GET:
00167                     struct sensor_packet state_ptr;
00168                     memcpy(&state_ptr, the_slave->getData(), sizeof(struct sensor_packet));
00169 
00170                     state_ptr.header.ptype = PacketType::DASHBOARD_RESPONSE;
00171 
00172                     send(&state_ptr, new_fd);
00173                     break;
00174             }
00175         }
00176     }
00177 }
00178 
00179 SlaveManager& BusServer::getSlaveManager() { return slave_manager; }
```

7.26 src/CO2Slave.cpp File Reference

```
#include "CO2Slave.h"
#include "packets.h"
```

Include dependency graph for CO2Slave.cpp:



7.27 CO2Slave.cpp

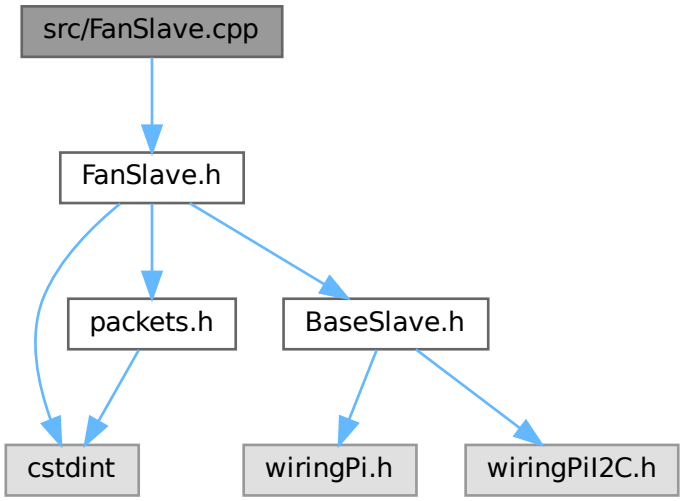
Go to the documentation of this file.

```
00001 #include "CO2Slave.h"
00002
00003 #include "packets.h"
00004
00005 CO2Slave::CO2Slave(uint8_t id, uint8_t i2c_address) : id(id) {
00006     state_packet.header.ptype = PacketType::DATA;
00007     state_packet.header.length = sizeof(struct sensor_packet_co2);
00008
00009     state_packet.data.rgb_light.metadata.sensor_id = id;
00010     state_packet.data.rgb_light.metadata.sensor_type = SensorType::CO2;
00011     command = 0x2003;
00012     uint8_t* command_ptr = (uint8_t*)&command;
00013
00014     wiringPiI2CWrite(fd, command_ptr[0]);
00015     wiringPiI2CWrite(fd, command_ptr[1]);
00016 }
00017
00018 const void* CO2Slave::getData() {
00019     command = 0x2008;
00020     uint8_t* command_ptr = (uint8_t*)&command;
00021     wiringPiI2CWrite(fd, command_ptr[0]);
00022     wiringPiI2CWrite(fd, command_ptr[1]);
00023     uint8_t* value_ptr = (uint8_t*)&state_packet.data.co2.value;
00024
00025     value_ptr[1] = wiringPiI2CRead(fd);
00026     value_ptr[0] = wiringPiI2CRead(fd);
00027     uint8_t empty = wiringPiI2CRead(fd);
00028     return &state_packet;
00029 }
00030
00031 bool CO2Slave::getStatus() { return power_state; }
00032
00033 int CO2Slave::getId() { return id; }
00034
00035 void CO2Slave::setData(void* data) { return; }
00036
00037 void CO2Slave::start(int i2c_fd) { fd = i2c_fd; }
00038
00039 void CO2Slave::stop() { fd = -1; }
```

7.28 src/FanSlave.cpp File Reference

```
#include "FanSlave.h"
```

Include dependency graph for FanSlave.cpp:



7.29 FanSlave.cpp

Go to the documentation of this file.

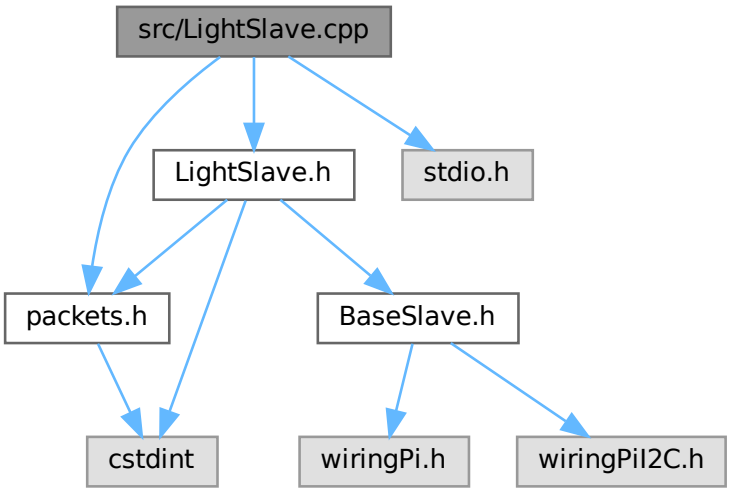
```
00001 #include "FanSlave.h"
00002
00003 FanSlave::FanSlave(uint8_t id, uint8_t i2c_address) : id(id), i2c_address(i2c_address) {}
00004
00005 const void* FanSlave::getData() { return &current_speed; }
00006
00007 bool FanSlave::getStatus() {
00008     if (current_speed == 0) {
00009         return false;
00010     } else {
00011         return true;
00012     }
00013 }
00014
00015 int FanSlave::getId() { return id; }
00016
00017 void FanSlave::setData(void* data) {
00018     current_speed = *(uint8_t*)data;
00019
00020     wiringPiI2CWrite(fd, current_speed);
00021 }
00022
00023 void FanSlave::start(int i2c_fd) { fd = i2c_fd; }
00024
00025 void FanSlave::stop() { fd = -1; }
```

7.30 src/LightSlave.cpp File Reference

```
#include "LightSlave.h"
#include <stdio.h>
```

```
#include "packets.h"
```

Include dependency graph for LightSlave.cpp:



7.31 LightSlave.cpp

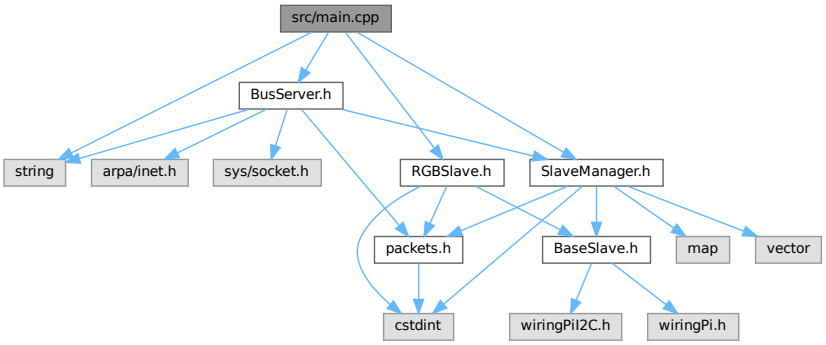
Go to the documentation of this file.

```
00001 #include "LightSlave.h"
00002
00003 #include <stdio.h>
00004
00005 #include "packets.h"
00006
00007 LightSlave::LightSlave(uint8_t id, uint8_t i2c_address) : id(id), i2c_address(i2c_address) {
00008     state_packet.header.ptype = PacketType::DATA;
00009     state_packet.header.length = sizeof(struct sensor_packet_light);
00010
00011     state_packet.data.light.metadata.sensor_id = id;
00012     state_packet.data.light.metadata.sensor_type = SensorType::LIGHT;
00013 }
00014
00015 const void* LightSlave::getData() {
00016     uint8_t value = wiringPiI2CRead(fd);
00017
00018     state_packet.data.light.target_state = value;
00019
00020     return &state_packet;
00021 }
00022
00023 bool LightSlave::getStatus() {
00024     getData();
00025     return state_packet.data.light.target_state;
00026 }
00027
00028 int LightSlave::getId() { return id; }
00029
00030 void LightSlave::setData(void* data) {
00031     uint8_t* value = (uint8_t*)data;
00032
00033     char command[256] = {0};
00034     snprintf(command, sizeof(command) - 1, "/usr/sbin/i2cset -y 1 0x%02x 0x%02x i", id, *value);
00035
00036     popen(command, "r");
00037 }
00038
00039 void LightSlave::start(int i2c_fd) { fd = i2c_fd; }
00040
00041 void LightSlave::stop() { fd = -1; }
```

7.32 src/main.cpp File Reference

```
#include <string>
#include "BusServer.h"
#include "RGBSlave.h"
#include "SlaveManager.h"
```

Include dependency graph for main.cpp:



Functions

- int main ()

7.32.1 Function Documentation

7.32.1.1 main()

```
int main ( )
```

Definition at line 7 of file main.cpp.

7.33 main.cpp

Go to the documentation of this file.

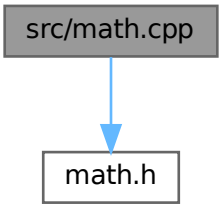
```
00001 #include <string>
00002
00003 #include "BusServer.h"
00004 #include "RGBSlave.h"
00005 #include "SlaveManager.h"
00006
00007 int main() {
00008     BusServer bus_server;
00009     bus_server.setup("0.0.0.0", 5000);
00010
00011     SlaveManager& slave_manager_ref = bus_server.getSlaveManager();
00012
00013     {
00014         int fd = wiringPiI2CSetup(0x69);
00015
00016         slave_manager_ref.createSlave(SensorType::RGB_LIGHT, 0x69);
00017         slave_manager_ref.getSlave(0x69)->start(fd);
00018     }
00019
00020     bus_server.start();
00021 }
```

7.34 src/math.cpp File Reference

Implementation of basic math operations.

```
#include "math.h"
```

Include dependency graph for math.cpp:



Functions

- int add (int a, int b)
  Adds two integers.
- int subtract (int a, int b)
  Subtracts two integers.

7.34.1 Detailed Description

Implementation of basic math operations.

Author

Daan Breur

Definition in file math.cpp.

7.34.2 Function Documentation

7.34.2.1 add()

```
int add (
            int a,
            int b )
```

Adds two integers.

Parameters

a	First integer.
b	Second integer.

Returns

The sum of a and b.

This function takes two integers as input and returns their sum.

Definition at line 16 of file math.cpp.

7.34.2.2 subtract()

```
int subtract (
            int a,
            int b )
```

Subtracts two integers.

Parameters

a	First integer.
b	Second integer.

Returns

The difference of a and b.

This function takes two integers as input and returns the result of subtracting b from a.

Definition at line 26 of file math.cpp.

7.35 math.cpp

Go to the documentation of this file.

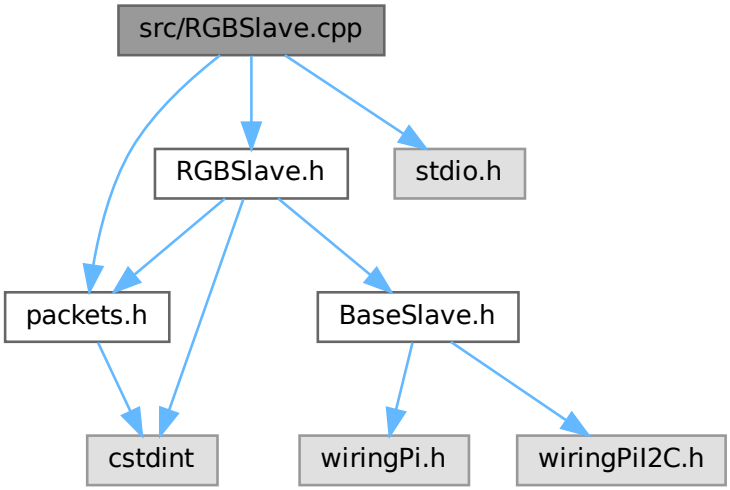
```
00001
00006 #include "math.h"
00007
00016 int add(int a, int b) { return a + b; }
00017
00026 int subtract(int a, int b) { return a - b; }
```

7.36 src/RGBSlave.cpp File Reference

```
#include "RGBSlave.h"
#include <stdio.h>
```

```
#include "packets.h"
```

Include dependency graph for RGBSlave.cpp:



7.37 RGBSlave.cpp

Go to the documentation of this file.

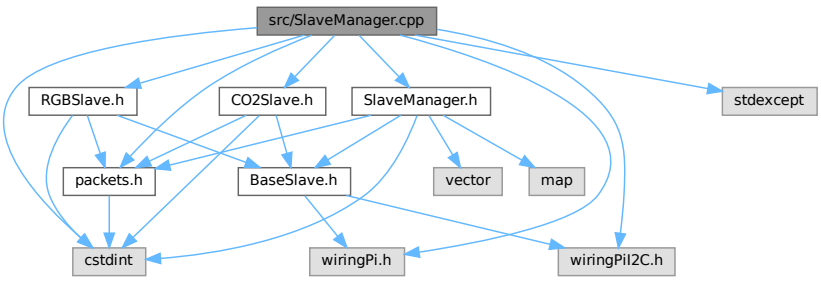
```
00001 #include "RGBSlave.h"
00002
00003 #include <stdio.h>
00004
00005 #include "packets.h"
00006
00007 RGBSlave::RGBSlave(uint8_t id, uint8_t i2c_address) : id(id), i2c_address(i2c_address) {
00008     state_packet.header.ptype = PacketType::DATA;
00009     state_packet.header.length = sizeof(struct sensor_packet_rgb_light);
00010
00011     state_packet.data.rgb_light.metadata.sensor_id = id;
00012     state_packet.data.rgb_light.metadata.sensor_type = SensorType::RGB_LIGHT;
00013 }
00014
00015 const void* RGBSlave::getData() {
00016     uint8_t r, g, b;
00017
00018     r = wiringPiI2CRead(fd);
00019     g = wiringPiI2CRead(fd);
00020     b = wiringPiI2CRead(fd);
00021
00022     color_state.R = r;
00023     color_state.G = g;
00024     color_state.B = b;
00025
00026     // state_packet.data.rgb_light.red_state = r;
00027     // state_packet.data.rgb_light.green_state = g;
00028     // state_packet.data.rgb_light.blue_state = b;
00029
00030     return &state_packet;
00031 }
00032
00033 bool RGBSlave::getStatus() {
00034     getData();
00035     if (color_state.R == 0 && color_state.G == 0 && color_state.B == 0) {
00036         return false;
00037     } else {
00038         return true;
00039     }
```

```
00040 }
00041
00042 int RGBSlave::getId() { return id; }
00043
00044 void RGBSlave::setData(void* data) {
00045     RGBData* rgb_data = (RGBData*)data;
00046     uint8_t r = rgb_data->R;
00047     uint8_t g = rgb_data->G;
00048     uint8_t b = rgb_data->B;
00049
00050     char command[256] = {0};
00051     snprintf(command, sizeof(command) - 1, "/usr/sbin/i2cset -y 1 0x%02x 0x%02x 0x%02x 0x%02x i",
00052              id, r, g, b);
00053
00054     state_packet.data.rgb_light.red_state = r;
00055     state_packet.data.rgb_light.green_state = g;
00056     state_packet.data.rgb_light.blue_state = b;
00057
00058     popen(command, "r");
00059
00060     // wiringPiI2CWriteBlockData(fd, 1, (uint8_t*)rgb_data, 3);
00061     // wiringPiI2CWrite(fd, g);
00062     // wiringPiI2CWrite(fd, b);
00063 }
00064
00065 void RGBSlave::start(int i2c_fd) { fd = i2c_fd; }
00066
00067 void RGBSlave::stop() { fd = -1; }
```

7.38 src/SlaveManager.cpp File Reference

```
#include "SlaveManager.h"
#include <wiringPi.h>
#include <wiringPiI2C.h>
#include <cstdint>
#include <stdexcept>
#include "CO2Slave.h"
#include "RGBSlave.h"
#include "packets.h"
```

Include dependency graph for SlaveManager.cpp:



Macros

- #define MASTER_ADDRESS 0x01

7.38.1 Macro Definition Documentation

7.38.1.1 MASTER_ADDRESS

```
#define MASTER_ADDRESS 0x01
```

Definition at line 13 of file SlaveManager.cpp.

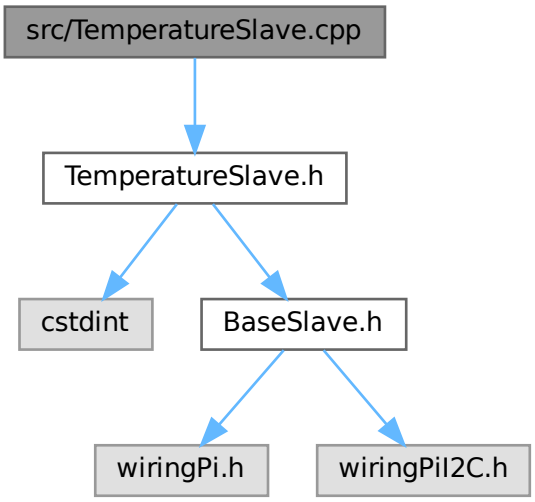
7.39 SlaveManager.cpp

Go to the documentation of this file.

```
00001 #include "SlaveManager.h"
00002
00003 #include <wiringPi.h>
00004 #include <wiringPiI2C.h>
00005
00006 #include <cstdint>
00007 #include <stdexcept>
00008
00009 #include "CO2Slave.h"
00010 #include "RGBSlave.h"
00011 #include "packets.h"
00012
00013 #define MASTER_ADDRESS 0x01
00014
00015 SlaveManager::SlaveManager() : address(MASTER_ADDRESS) {}
00016
00017 void SlaveManager::initialize() {
00018     // maybe need setup not sure.
00019
00020     i2c_fd = wiringPiI2CSetup(address);
00021     if (-1 == i2c_fd) {
00022         throw std::runtime_error("I2C setup failed");
00023     } else {
00024         printf("I2C setup succesful");
00025     };
00026 }
00027
00028 void SlaveManager::ledControl(uint8_t led_number, uint8_t led_state) {}
00029
00030 void SlaveManager::createSlave(SensorType type, int i2c_address) {
00031     BaseSlave* newSlave = nullptr;
00032     switch (type) {
00033         case SensorType::CO2:
00034             newSlave = new CO2Slave(i2c_address, i2c_address);
00035             break;
00036         case SensorType::RGB_LIGHT:
00037             newSlave = new RGBSlave(i2c_address, i2c_address);
00038             break;
00039         // case /* door */:
00040         //     newSlave = new DoorSlave(id, i2c_address);
00041         //     break;
00042         default:
00043             return;  // Invalid type
00044     }
00045     slaves[i2c_address] = newSlave;
00046 }
00047 void SlaveManager::deleteSlave(uint8_t id) {
00048     if (nullptr != slaves[id]) {
00049         /*
00050          * Disconnect I2C device from bus
00051          */
00052         delete slaves[id];
00053     }
00054 }
00055
00056 BaseSlave* SlaveManager::getSlave(int id) { return slaves[id]; }
```

7.40 src/TemperatureSlave.cpp File Reference

`#include "TemperatureSlave.h"`
Include dependency graph for TemperatureSlave.cpp:



7.41 TemperatureSlave.cpp

Go to the documentation of this file.

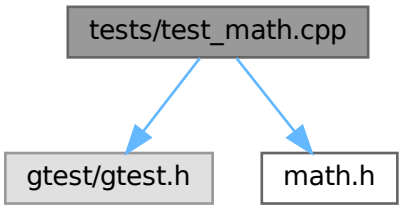
```
00001 #include "TemperatureSlave.h"
00002
00003 TemperatureSlave::TemperatureSlave(uint8_t id, uint8_t i2c_address)
00004     : id(id), i2c_address(i2c_address) {}
00005
00006 void* TemperatureSlave::getData() {
00007     int temp;
00008
00009     temp = wiringPiI2CRead(fd);
00010
00011     return;
00012 }
00013
00014 bool TemperatureSlave::getStatus() {
00015     getData();
00016     if (temperature.Temp == 0) {
00017         return false;
00018     } else {
00019         return true;
00020     }
00021 }
00022
00023 int TemperatureSlave::getId() { return id; }
00024
00025 void TemperatureSlave::setData(void* data) {
00026     TemperatureData* temp_data = (TemperatureData*)data;
00027     int temp = temp_data->Temp;
00028
00029     wiringPiI2CWrite(fd, temp);
00030 }
00031
00032 void TemperatureSlave::start(int i2c_fd) { fd = i2c_fd; }
00033
00034 void TemperatureSlave::stop() { fd = -1; }
```

7.42 tests/test_math.cpp File Reference

Unit tests for mathematical operations using Google Test framework.

```
#include <gtest/gtest.h>
#include "math.h"
```

Include dependency graph for test_math.cpp:



Functions

- TEST (MathTest, Add)
- TEST (MathTest, Subtract)
- TEST (MathTest, SubtractNegative)

7.42.1 Detailed Description

Unit tests for mathematical operations using Google Test framework.

This file contains test cases for verifying the correctness of functions defined in the "math.h" header. The tests ensure that the mathematical operations behave as expected under various conditions.

Test MathTest.Add

- Verifies that the `add` function correctly computes the sum of two integers.
- Example: `add(2, 3)` should return `5`.

Test MathTest.Subtract

- Verifies that the `subtract` function correctly computes the difference between two integers.
- Examples:
  - `subtract(10, 3)` should return `7`.
  - `subtract(9, 3)` should return `6`.

Test MathTest.SubtractNegative

- Verifies that the `subtract` function handles subtraction with negative integers correctly.
- Example: `subtract(10, -3)` should return `13`.

Definition in file test_math.cpp.

7.42.2 Function Documentation

7.42.2.1 TEST() [1/3]

```
TEST (
            MathTest ,
            Add  )
```

Definition at line 29 of file test_math.cpp.

7.42.2.2 TEST() [2/3]

```
TEST (
            MathTest ,
            Subtract  )
```

Definition at line 31 of file test_math.cpp.

7.42.2.3 TEST() [3/3]

```
TEST (
            MathTest ,
            SubtractNegative  )
```

Definition at line 36 of file test_math.cpp.

7.43 test_math.cpp

Go to the documentation of this file.

```
00001
00025 #include <gtest/gtest.h>
00026
00027 #include "math.h"
00028
00029 TEST(MathTest, Add) { EXPECT_EQ(add(2, 3), 5); }
00030
00031 TEST(MathTest, Subtract) {
00032     EXPECT_EQ(subtract(10, 3), 7);
00033     EXPECT_EQ(subtract(9, 3), 6);
00034 }
00035
00036 TEST(MathTest, SubtractNegative) { EXPECT_EQ(subtract(10, -3), 13); }
```

Index